



Momentum Across Market Regimes – A Machine Learning Approach

By

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- **Proofreading and clarity.** Grammar correction, copy-editing, and suggestions for improving the readability of the author's drafts.
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Management Summary

Cross-sectional momentum, the tendency for recent stock-market winners to keep winning and losers to keep losing, is one of the most profitable anomalies in finance and one of the most fragile. Momentum portfolios suffered losses exceeding 50% during the 2009 market rebound and have crashed repeatedly around regime transitions. The standard responses scale exposure during stress or blend horizon weights, leaving the selection rule itself unconditional. This thesis asks a different question: how does the cross-sectional structure of momentum reorganize between calm and panic regimes, and what mechanism drives that reorganization?

The empirical framework has two stages. A Bayesian two-state Hidden Markov Model, estimated by Gibbs sampling on four monthly macrofinancial stress indicators, produces a continuous filtered probability of market panic that updates each month. This regime signal is then combined with twelve momentum horizons inside a cross-sectional stock-selection model (an XGBoost gradient-boosted regressor, benchmarked against a deterministic formula and a logistic regression). Stocks are ranked monthly and the top and bottom deciles enter a value-weighted long-short portfolio. All models are estimated on 1990–2010 U.S. equities and evaluated out-of-sample on the 167-month 2011–2024 window, with international replication on UK and Japanese equities.

The main finding concerns *which stocks* the strategy holds across regimes, not *how much* it holds. Prior crash-management work adjusts the strategy’s exposure during stress while leaving its selection rule unchanged; this thesis shows that the selection rule itself reorganizes between regimes. In calm the assignment is conventional: long past winners. In panic the model inverts that assignment: the long leg is rebuilt from the most beaten-down names, and the long-short portfolio remains fully invested throughout. Composition and exposure are complementary channels rather than substitutes, and a hybrid that adds an exposure overlay on top of this selection rule is left to future work. The thesis contributes both the methodological combination that makes this kind of regime-conditioned selection possible and an empirical characterization of what the reorganization looks like at the level of individual cluster z-scores (Section 5.2.3).

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1 Introduction

This thesis applies machine learning to study the behavior of cross-sectional momentum¹ across market regimes. The aim is to better understand this anomaly conditional on the market regime, not to propose a strategy that outperforms the market on aggregate.

Cross-sectional momentum is one of the most robust anomalies in empirical finance and one of the most fragile. This fragility motivates a substantial literature on momentum-crash management, but existing responses share a structural assumption. The regime signal enters only to scale or re-weight a fixed scoring rule, never as a feature inside the rule itself. Barroso and Santa-Clara (2015) target constant volatility during stress, while Cooper et al. (2004) treat regimes as a binary up-or-down variable and Goulding et al. (2023) blend fast and slow horizon weights. This leaves unexamined how the full momentum term structure and the regime signal jointly determine which stocks the strategy selects in each market state.

This thesis examines that possibility. The regime signal acts as a *context variable*: it carries no direct return-predictive content, but conditions how a nonlinear cross-sectional model uses the momentum term structure in a given market state.

Research question

The thesis asks how the cross-sectional structure of momentum changes between calm and panic regimes, and what mechanism drives that change.

Approach

The question is answered in two steps. First, a Bayesian Hidden Markov Model (HMM) is estimated on four monthly macrofinancial stress indicators (drawdown, return dispersion, relative market participation, and credit spread), producing a filtered probability of market panic that updates monthly. This regime signal is combined with the twelve momentum horizons in cross-sectional stock-selection models (deterministic formula, logistic regression, XGBoost ensemble), each evaluated out-of-sample on 2011–2024 U.S. equities in long-short construction. Second, the chosen model’s selection rule is opened up month by month with a per-feature contribution analysis and a clustering of long-leg cross-sectional z-curves, identifying the role the regime signal plays in selection.

Headline findings

On 2011–2024 U.S. equities, the XGBoost ensemble on the regime probability and the twelve momentum horizons produces a six-factor alpha of 24.1% ($t = 4.81$). A linear baseline on identical features collapses, and removing the regime probability from XGBoost alone cuts its Sharpe by more than 60%. The mechanism is that the cross-section of

¹Cross-sectional momentum ranks stocks against each other at each point in time, going long the relative winners and short-selling the relative losers within the same universe (Jegadeesh and Titman, 1993).

momentum reorganizes systematically across regimes. In panic, the regime signal flips the model’s selection direction while keeping the long-short portfolio fully invested. The long leg holds stocks not top-ranked at any individual momentum horizon, and the long-leg market beta stays above the short-leg beta in every regime, in contrast to the crash mechanism documented for unconditional momentum. Cluster analysis recovers four recurring selection modes that trace a single market cycle.

Contributions

The thesis makes two contributions to the momentum literature. First, the methodological combination of a continuous probabilistic regime filter (the Bayesian HMM filtered probability of market panic) with a nonlinear cross-sectional model (XGBoost) is itself novel relative to the linear and rank-based regime-conditional rules in the prior literature. The combination enables a class of regime-conditioned selection not documented in the prior momentum literature, including a calm-panic direction flip already documented descriptively by [Cooper et al. \(2004\)](#) but here implemented concretely as a single nonlinear model with the long-short portfolio fully invested throughout. Second, the empirical analysis characterizes how that mechanism manifests at a granularity the prior literature does not provide, decomposing XGB’s monthly picks into four recurring selection modes and showing that the regime signal acts as an irreducibly nonlinear routing variable for the momentum term structure.

Outline

Section 2 (Literature Review) covers momentum’s horizon structure, momentum crashes and exposure-based risk management, regime models in asset pricing, and machine learning in cross-sectional finance. Section 3 (Methodology) develops the two-stage framework, including the Bayesian HMM specification and Gibbs sampling estimation, and the cross-sectional stock-selection methods. Section 4 (Data) describes the U.S. equity sample, the four-pass HMM feature selection pipeline, and the cross-sectional features. Section 5 (Results) presents the out-of-sample results, the cross-sectional z-score and cluster analysis that documents the regime-momentum mechanism, robustness and risk-aversion analyses, and international replication. Section 6 (Summary, Conclusions and Recommendations) summarizes the contributions, discusses limitations, and suggests directions for future research.

2 Literature Review

2.1 The Momentum Anomaly and Its Horizon Structure

Jegadeesh and Titman (1993) established cross-sectional momentum as a pervasive feature of equity markets. Portfolios that buy past winners and sell past losers earn significant returns over 3-to-12-month holding periods. They examine sixteen formation-and-holding-period combinations from three to twelve months, with the twelve-month formation period emerging as the most profitable. The strategy has since been replicated across international markets, asset classes, and sample periods (Jegadeesh and Titman, 2001; Asness et al., 2013).

The economic mechanism behind momentum remains debated. Behavioral explanations include gradual information diffusion (Hong and Stein, 1999) and the joint conservatism-representativeness mechanism of Barberis et al. (1998), while risk-based explanations link momentum to time-varying systematic risk. This thesis does not take a position on the underlying mechanism but examines the empirical regularity that momentum’s cross-sectional structure varies with market conditions.

A defining feature of momentum is its sensitivity to the lookback horizon. Short lookbacks of one to three months capture short-term reversal effects (Jegadeesh, 1990; Lehmann, 1990), intermediate horizons of seven to twelve months prior to formation capture continuation (Novy-Marx, 2012), and lookbacks beyond twelve months reflect long-term mean reversion (De Bondt and Thaler, 1985). Lee and Swaminathan (2000) document a momentum lifecycle in which the same signal evolves from continuation to reversal as information ages. Together these findings suggest that momentum is a family of horizon-specific patterns whose relative strength varies with market conditions, not a single signal. The framework here tests this directly by exposing all twelve monthly horizons jointly to the cross-sectional model rather than committing to a single lookback *ex ante*.

2.2 Momentum Crashes and Exposure-Based Solutions

Despite its long-run profitability, momentum is prone to severe crashes. Daniel and Moskowitz (2016) document that momentum portfolios suffered catastrophic losses during the 1932 and 2009 market rebounds, and show that these crashes are predictable. When the market has recently declined, past losers tend to be high-beta stocks that rally disproportionately during recoveries, while past winners are low-beta defensive stocks that lag. In economic terms, momentum is a bet that the recent regime persists. It earns the continuation premium when the regime holds, and crashes at regime transitions, when the cross-sectional ordering inherited from the prior regime becomes the wrong bet for the new

one. Barroso and Santa-Clara (2015) propose a simple risk-management response, scaling momentum exposure inversely with the strategy’s recent realized volatility, a constant-volatility overlay that substantially raises the Sharpe ratio. Bianchi et al. (2022) pursue a related approach by constructing a crash indicator from the joint time variation of conditional volatility and skewness, again operating on the strategy’s exposure rather than its composition. This thesis explores a complementary composition-side route: rather than scaling exposure during stress, the model optimizes on which stocks populate each leg conditional on the regime. As a consequence, the cross-section is re-ranked across regimes and the long-leg market beta remains above the short-leg beta in every regime (Appendix H.5).

The liquidity literature provides a theoretical basis for why momentum crashes reverse. Margin spirals (Brunnermeier and Pedersen, 2009) and institutional flow pressure (Vayanos and Woolley, 2013) create temporary price dislocations during stress that revert when selling pressure subsides, the mechanism central to the regime-momentum interpretation developed below.

2.3 Regime Models and Regime-Conditional Asset Pricing

Hidden Markov Models have a long history in econometrics. The probabilistic machinery dates to Baum et al. (1970); Hamilton (1989) introduced the regime-switching tradition with a business-cycle application to U.S. GNP growth, and Bayesian estimation via Gibbs sampling was developed by Albert and Chib (1993) with the multivariate-emission generalization now standard (Frühwirth-Schnatter, 2006). Following Hamilton, this thesis fits the regime model to macrofinancial stress indicators rather than to returns themselves.

A separate literature asks whether asset-pricing anomalies are themselves regime-conditional. Cooper et al. (2004) establish that momentum profits depend on the prior market state, with strong continuation following up-markets and weaker or reversed performance following down-markets. More recent work integrates regime estimation into portfolio construction, for example Shu and Mulvey (2024), who feed sparse-jump-model factor regimes into a Black-Litterman allocation.

The most directly related work is Goulding et al. (2023) (hereafter GHM), who classify each month into one of four market cycles (Bull, Correction, Bear, Rebound) from the trailing one-month and rolling twelve-month market returns and blend the two momentum horizons via a cycle-specific weight $\alpha \cdot mom_1 + (1 - \alpha) \cdot mom_{12}$. The framework here generalizes that approach along three axes: from a deterministic four-state classification based on trailing returns to a continuous panic probability from a Bayesian HMM on real-time stress indicators; from two horizons to the full twelve-horizon term structure; and from a parametric linear blend to a learned nonlinear combination. GHM’s strategies are a time-series optimization of a single market index rather than a cross-sectional long-short

of individual stocks, so a direct performance comparison is not appropriate.

The unifying feature of this subsection is that the regime variable enters portfolio decisions through a hand-coded rule, threshold, or parametric weight, never as a continuous learned feature. This thesis instead supplies the probability of being in a calm/panic regime as a feature to a nonlinear cross-sectional model, which learns how to use it to optimize portfolio construction.

2.4 Machine Learning in Cross-Sectional Asset Pricing

The use of machine learning methods for cross-sectional stock selection has grown rapidly. [Gu et al. \(2020\)](#) compare a wide range of methods on a kitchen-sink characteristic set including multiple momentum horizons, finding that tree-based models and neural networks substantially outperform linear models by capturing nonlinear interactions among firm characteristics. Gradient-boosted trees ([Chen and Guestrin, 2016](#)) have proven particularly effective for tabular financial data because they handle missing values natively, capture nonlinear relationships and interactions without explicit specification, and are robust to irrelevant features. A central challenge in applying these methods to asset pricing is interpretability. This thesis addresses it on two fronts: SHAP values ([Lundberg and Lee, 2017](#); [Lundberg et al., 2020](#)) attribute each prediction to feature-level contributions, and a clustering of the long leg’s monthly cross-sectional z-curves into recurring selection modes characterizes what the model selects in each regime rather than only which features matter.

The closest methodological precedent is [Beckmeyer and Wiedemann \(2025\)](#), who use machine learning to learn data-driven weights across past daily returns for a single momentum signal, constructing a dynamically reweighted strategy that outperforms the conventional twelve-minus-one specification. The framework here differs in two respects. First, the weighting is learned not over the daily history of one signal but over a vector of twelve monthly horizons. Second, the weighting is learned *conditional on a regime context variable*, in contrast to the unconditional weighting in [Beckmeyer and Wiedemann \(2025\)](#), which averages across calm and panic regimes. Isolating the regime-conditioning role is this framework’s primary contribution, and the thesis tests whether disaggregating that average reveals a structural reorganization of the term structure between states.

Despite these advances, to the best of my knowledge no published study examines how the full term structure of cross-sectional momentum predictability reorganizes across regimes, or whether capturing this reorganization requires nonlinear modeling.

3 Methodology

This chapter combines ideas and statistical methods from the literatures reviewed in Section 2 into a two-stage framework that tests the research question posed in Section 1. Stage 1 estimates a Bayesian two-state Hidden Markov Model on four macrofinancial stress indicators, producing a continuous probability of the panic state. Stage 2 combines that probability with cross-sectional momentum inside a series of stock-selection models.

3.1 Stage 1: Regime Classification via Hidden Markov Model

3.1.1 Model Specification

This thesis employs a two-state ($K = 2$) Bayesian Hidden Markov Model to classify the market into latent regimes². The hidden state $s_t \in \{0, 1\}$ represents the unobserved regime at month t , where $s_t = 0$ denotes the *calm* regime and $s_t = 1$ denotes the *panic* regime. The labels *calm* and *panic* reflect the economic nature of the chosen stress features, which are diagnostic of market stress rather than trend direction, business-cycle phase, or volatility alone. An HMM fitted to a different feature set would identify different regimes. As emphasized by Hamilton (1989), market regimes are not directly observable. An investor cannot know with certainty whether the market is currently in a calm or panic state, and classifications that look obvious in retrospect are genuinely difficult to identify in real time. The HMM formalizes this uncertainty by treating regimes as hidden states that must be inferred probabilistically from observable market data.

Observation equation

Conditional on the regime, the vector of standardized features³ $\mathbf{z}_t \in \mathbb{R}^D$ is drawn from a regime-specific multivariate normal distribution. We assume that, conditional on the regime $s_t = k$, $\mathbf{z}_t$ is normally distributed with mean vector $\boldsymbol{\mu}_k$ and covariance matrix $\boldsymbol{\Sigma}_k$ for $k \in \{0, 1\}$ (calm and panic). Writing $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ for the multivariate normal distribution with the indicated mean and covariance, and using the shorthand $\mathbf{z}_{1:t} = (\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_t)$ for the history of observations through month t ,

$$(\mathbf{z}_t \mid s_t = 0) \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0), \quad (\mathbf{z}_t \mid s_t = 1) \sim \mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1). \quad (1)$$

Each regime is characterized by its own mean vector $\boldsymbol{\mu}_k \in \mathbb{R}^D$ and covariance matrix $\boldsymbol{\Sigma}_k \in \mathbb{R}^{D \times D}$, where $k \in \{0 \text{ (calm)}, 1 \text{ (panic)}\}$. The multivariate specification allows the model to capture not only level differences in individual features across regimes, but also changes

²The choice of $K = 2$ is validated empirically. Appendix G.4 shows that increasing to $K = 3, 4$, or 5 states does not improve downstream portfolio performance and introduces estimation instability.

³ $\mathbb{R}^D$ denotes the D -dimensional space of real-valued vectors; the D stress indicators at month t are stacked into a single point in this space so the HMM can model them jointly. The value of D is set by feature selection (Section 4.1).

in the correlation structure. For instance, stress features may become more correlated during panic periods. The Gaussian assumption is standard in the HMM literature. Appendix H.8 verifies its adequacy by comparing it to heavier-tailed multivariate Student- t emissions.

Transition equation

The evolution of the hidden state follows a first-order Markov chain with a 2×2 transition matrix:

$$\mathbf{P} = \begin{pmatrix} P_{00} & P_{01} \\ P_{10} & P_{11} \end{pmatrix}, \quad (2)$$

where $P_{ij} = P(s_t = j \mid s_{t-1} = i)$ and each row sums to one. The diagonal elements P_{00} (calm persistence) and P_{11} (panic persistence) govern how likely each regime is to continue into the next month: the expected duration of regime k is $1/(1 - P_{kk})$ months, so a persistence of 0.95 corresponds to a regime that lasts roughly 20 months on average.

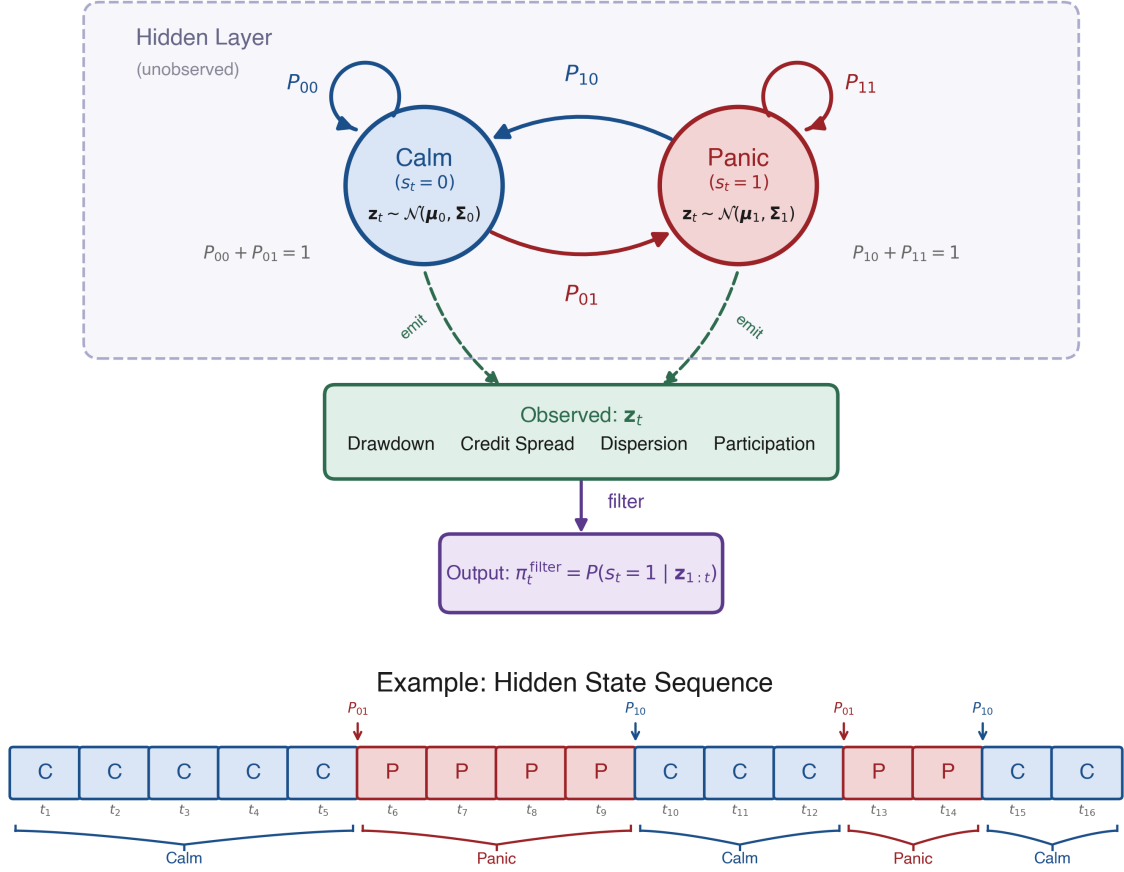


Figure 1: Architecture of the two-state Hidden Markov Model. Hidden states (Calm/Panic) evolve via transition probabilities P_{ij} and emit the observation vector $\mathbf{z}_t \in \mathbb{R}^D$. The forward filter inverts this process to produce $\pi_t^{\text{filter}} = P(s_t = 1 | \mathbf{z}_{1:t})$. The lower panel shows an example state sequence illustrating regime persistence and abrupt transitions.

3.1.2 Bayesian Estimation via Gibbs Sampling

The standard frequentist alternative is the Baum–Welch algorithm (Baum et al., 1970), a special case of Expectation-Maximization (Dempster et al., 1977), which finds maximum likelihood point estimates.

Maximum likelihood estimation of regime-switching HMMs has several known shortcomings for the present application. The likelihood surface is multimodal, with at minimum two equivalent global optima arising from the invariance of the likelihood under state relabeling. Additionally, expectation maximization can converge to degenerate solutions in which one regime collapses around a single observation (Frühwirth-Schnatter, 2006). Both problems are exacerbated for rare regimes. The asymptotic standard errors that frequentist inference relies on become unreliable when few observations support the rare-

state estimates, and the standard regularity conditions can fail outright: the persistence parameters P_{00} and P_{11} are unidentified when the two regimes' means and covariances coincide, and the information matrix becomes singular at that point (Hamilton, 1989, footnote 12). The Bayesian approach addresses these issues directly. Posterior sampling integrates over parameter uncertainty rather than relying on a point estimate. Conjugate priors regularize rare-regime estimates with explicit, defensible weights (Section 3.1.3). And convergence diagnostics across multiple chains (Appendix J) diagnose multimodality that would otherwise go unnoticed under a single EM run.

The Bayesian approach instead draws from the joint posterior:

$$p(\{s_t\}_{t=1}^T, \{\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}_{k=0}^1, \mathbf{P} \mid \mathbf{z}_{1:T}).$$

The states and the parameters are coupled: we can sample from the distribution of one block given the other, but not from either block on its own. Gibbs sampling (Albert and Chib, 1993; Hoff, 2009), a Markov chain Monte Carlo (MCMC) method, exploits exactly that structure by alternating between three conditional updates over the state path, the emission parameters, and the transition matrix. Conjugate priors ensure that each block can be sampled exactly from its conditional posterior.

3.1.3 Prior Specification

Emission parameters

Before seeing the data, neither regime should be forced to have extreme values in any feature; the model should learn the characteristics of calm and panic states from the data. This is encoded by a conjugate Normal-Inverse-Wishart (NIW) prior on each regime's mean vector and covariance matrix:⁴

$$\boldsymbol{\Sigma}_k \sim \mathcal{IW}(\nu_0, \boldsymbol{\Psi}_0), \quad \boldsymbol{\mu}_k \mid \boldsymbol{\Sigma}_k \sim \mathcal{N}\left(\mathbf{m}_0, \frac{\boldsymbol{\Sigma}_k}{\kappa_0}\right). \quad (3)$$

The hyperparameters are chosen as

$$\mathbf{m}_0 = \mathbf{0}, \quad \kappa_0 = 0.01, \quad \nu_0 = D + 2, \quad \boldsymbol{\Psi}_0 = \mathbf{I}_D(\nu_0 - D - 1), \quad (4)$$

where D is the number of selected features. After observing data, these prior hyperparameters update to posterior counterparts κ_n , $\mathbf{m}_n$, ν_n , and $\boldsymbol{\Psi}_n$ (Equation 9). Each hyperparameter controls a different aspect of the prior belief:

- $\mathbf{m}_0$ is the prior guess for each regime's mean vector. $\mathbf{m}_0 = \mathbf{0}$ is the natural prior

⁴The Inverse-Wishart is a probability distribution over positive-definite matrices, making it the natural prior for a covariance matrix. Every sample is guaranteed to be symmetric and positive-definite. The Normal-Inverse-Wishart combines this with a conditional Normal prior on the mean.

given that the features are standardized. Without data, neither regime is presumed to center away from zero.

- κ_0 controls how strongly the prior mean $\mathbf{m}_0$ is trusted relative to the data. The very small value $\kappa_0 = 0.01$ makes this prior nearly uninformative, so the posterior mean is driven overwhelmingly by the observed data.
- ν_0 sets the Inverse-Wishart prior’s degrees of freedom, interpretable as the number of pseudo-observations the prior is worth. The value $\nu_0 = D + 2$ is the smallest that yields a proper prior with finite mean; a larger choice tightens the prior around $\mathbf{I}_D$.
- Ψ_0 is the scale matrix of the Inverse-Wishart prior. It is calibrated so that the prior expected covariance equals the identity matrix, $\mathbb{E}[\Sigma_k] = \Psi_0/(\nu_0 - D - 1) = \mathbf{I}_D$, implying unit marginal variance and no cross-feature correlation before observing the data.

Transition matrix

Regimes should be persistent rather than switching every few months. This is encoded by assigning each row of the transition matrix an independent Dirichlet prior whose mass concentrates on the diagonal:

$$\mathbf{P}_{0,\cdot} \sim \text{Dir}(9, 1), \quad \mathbf{P}_{1,\cdot} \sim \text{Dir}(1, 9). \quad (5)$$

Equivalently, in matrix form,

$$\boldsymbol{\alpha}_{\text{dir}} = \begin{pmatrix} 9 & 1 \\ 1 & 9 \end{pmatrix}.$$

This prior favors persistence in both regimes. Without data, the prior mean transition matrix is

$$\mathbb{E}[\mathbf{P}] = \begin{pmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{pmatrix}. \quad (6)$$

The prior expects each regime to continue with probability 0.9, corresponding to an expected duration of roughly 10 months. It is informative enough to discourage pathological solutions but weak enough to be overridden by the data.

3.1.4 Gibbs Sampling Algorithm

Initialization

The Gibbs sampler requires starting values for the emission parameters and the transition matrix. A crude binary partition based on a single stress indicator is used:⁵ months above

⁵The choice of initialization has negligible impact on the posterior. The sampler is run for 2,000 iterations with 500 discarded as burn-in, and convergence diagnostics (Section 4.3, Appendix J) confirm that the chain mixes well regardless of the starting partition.

the indicator’s median (the more stressed half) are initially assigned to the panic state ($s_t = 1$), and the rest to the calm state ($s_t = 0$). The initial emission parameters are then computed from each group:

$$\boldsymbol{\mu}_k^{(0)} = \frac{1}{n_k} \sum_{t: s_t=k} \mathbf{z}_t, \quad \boldsymbol{\Sigma}_k^{(0)} = \text{Cov}(\{\mathbf{z}_t : s_t = k\}),$$

where n_k is the number of months assigned to regime k . The initial transition matrix is set to the prior mean, $\mathbf{P}^{(0)} = \mathbb{E}[\mathbf{P}]$ (Equation 6). These initial parameters give the first forward pass (described below) emission densities that can distinguish the two states.

Iterative updates

Each iteration cycles through three conditional updates: the latent state path $\{s_t\}_{t=1}^T$, the emission parameters $(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$, and the transition matrix $\mathbf{P}$. Each block is sampled from its full conditional distribution, with the other two blocks held fixed at their current values; for Steps 2 and 3 this distribution takes the conjugate Bayes form posterior $\propto \text{likelihood} \times \text{prior}$, while Step 1 (FFBS) samples the state path jointly from its full conditional via the forward-filtering backward-sampling recursion described below. The likelihood factor that appears throughout is the Gaussian emission density $f(\mathbf{z}_t \mid s_t = k) = \mathcal{N}(\mathbf{z}_t; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$, the plausibility of the observed feature vector $\mathbf{z}_t$ under regime k . The cycle is summarised graphically below; the steps that follow derive each update in turn.

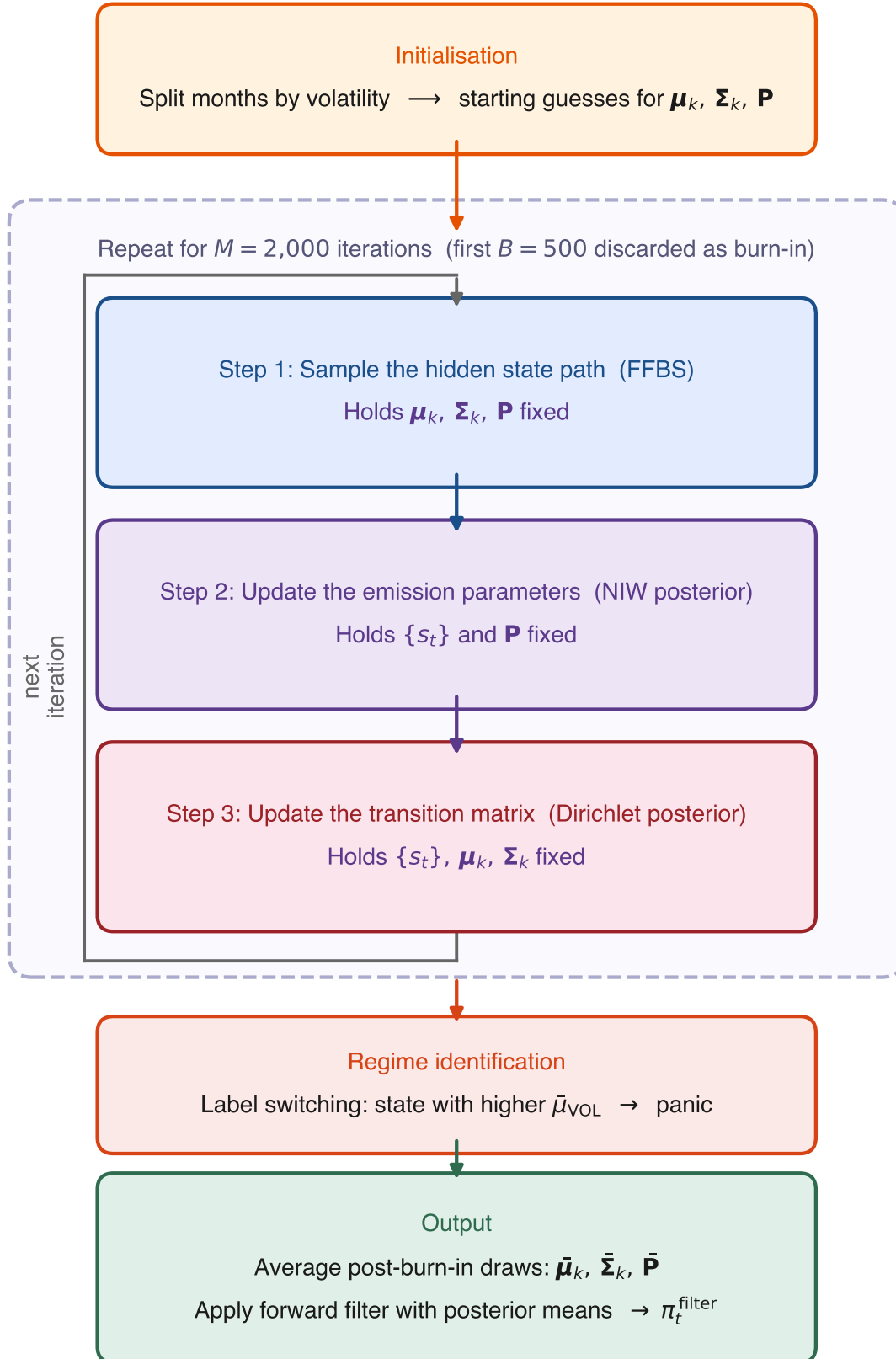


Figure 2: Flowchart of the Gibbs sampler.

Step 1: Sampling the hidden state path via FFBS

The full latent regime sequence $(s_1, \dots, s_T)$ is sampled using Forward-Filtering Backward-Sampling (FFBS; see [Frühwirth-Schnatter, 2006](#), ch. 11), with the current emission parameters and transition matrix held fixed.

The forward pass (the *forward filter*) computes the filtered probability

$$\alpha_t(k) = P(s_t = k \mid \mathbf{z}_{1:t}),$$

the probability of being in state k at time t given all observations so far. $\alpha_t(k)$ is built in two sub-steps. First, a *prediction step* asks: before seeing this month’s data, how likely is each state? Since the previous state is unknown, the algorithm sums over both possibilities, weighting each by its filtered probability from the previous month and the transition probability:

$$\alpha_{t|t-1}(k) = \alpha_{t-1}(0) P_{0k} + \alpha_{t-1}(1) P_{1k}.$$

Second, an *update step* revises this prediction once the current observation $\mathbf{z}_t$ arrives, using Bayes’ rule. If the observed features look more like a panic month than a calm month, the probability of panic increases. Combining the two steps yields the recursion

$$\alpha_t(k) = \frac{f(\mathbf{z}_t \mid s_t = k) \alpha_{t|t-1}(k)}{f(\mathbf{z}_t \mid s_t = 0) \alpha_{t|t-1}(0) + f(\mathbf{z}_t \mid s_t = 1) \alpha_{t|t-1}(1)}, \quad (7)$$

where $f(\mathbf{z}_t \mid s_t = k) = \mathcal{N}(\mathbf{z}_t; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ is the Gaussian emission density.⁶ The recursion is initialized with a uniform initial state distribution $P(s_1 = k) = 1/2$, reflecting no prior preference for either regime. For the first observation this gives

$$\alpha_1(k) \propto f(\mathbf{z}_1 \mid s_1 = k).$$

The filtered probabilities $\alpha_t(k)$ are marginals: they describe each month’s regime in isolation, not how consecutive months connect. Sampling each s_t independently from its α_t would ignore the transition matrix and could produce paths that flip regimes from one month to the next, in conflict with the strongly persistent $\mathbf{P}$. A second, backward pass is therefore needed to draw a full state path that respects $\mathbf{P}$.

The backward pass proceeds in two steps:

⁶In practice, these densities can be extremely small (or large) for multi-dimensional observations, leading to numerical underflow. The implementation avoids this by working in log-space. Log-likelihoods $\log f(\mathbf{z}_t \mid s_t = k)$ are computed directly, and the log-sum-exp identity $\log(e^a + e^b) = \max(a, b) + \log(1 + e^{-|a-b|})$ is used whenever probabilities must be normalized.

Terminal state

At $t = T$, the filtered probability already conditions on all observations, so the terminal state is drawn directly:

$$s_T \sim \text{Cat}(\alpha_T(0), \alpha_T(1)),$$

that is, the algorithm draws $s_T = 0$ (calm) with probability $\alpha_T(0)$ and $s_T = 1$ (panic) with probability $\alpha_T(1)$.⁷

Backward recursion

For each earlier time step $t = T - 1, T - 2, \dots, 1$, the algorithm chooses a state consistent with two pieces of information: (i) the filtered belief $\alpha_t(k)$ from the forward pass, and (ii) the already-sampled future state s_{t+1} , which constrains the current state via the transition probability $P_{k,s_{t+1}}$. Combining these two terms and normalizing gives:

$$P(s_t = k \mid s_{t+1}, \mathbf{z}_{1:t}) = \frac{\alpha_t(k) P_{k,s_{t+1}}}{\alpha_t(0) P_{0,s_{t+1}} + \alpha_t(1) P_{1,s_{t+1}}}. \quad (8)$$

For example, if the forward filter assigns high probability to calm at time t ($\alpha_t(0)$ is large) and the already-sampled s_{t+1} is also calm (so $P_{0,0}$ is large), then the product $\alpha_t(0) P_{0,0}$ dwarfs $\alpha_t(1) P_{1,0}$, the conditional probability of $s_t = 0$ approaches one, and the algorithm very likely samples $s_t = 0$.

The output of the full FFBS procedure is one draw from the posterior over state paths. Across Gibbs iterations, different paths are sampled, exploring the full posterior uncertainty.

The backward pass is used only during training: the sampled state path it produces feeds the conjugate Gibbs updates of the emission and transition parameters. The trading signal π_t^{filter} relies solely on the forward filter.⁸

Step 2: Updating the emission parameters

Given the sampled state path, each regime k has n_k observations assigned to it. From these, two summary statistics are computed:

$$\bar{\mathbf{x}}_k = \frac{1}{n_k} \sum_{t: s_t=k} \mathbf{z}_t, \quad \mathbf{S}_k = \sum_{t: s_t=k} (\mathbf{z}_t - \bar{\mathbf{x}}_k)(\mathbf{z}_t - \bar{\mathbf{x}}_k)^\top.$$

⁷Cat denotes the categorical distribution, the standard notation for a single draw from a discrete probability vector. With two states it reduces to a weighted coin flip.

⁸The forward filter uses only observations $\mathbf{z}_{1:t}$ (past and present) at each month t , never future months $\mathbf{z}_{t+1:T}$, so no look-ahead information leaks into the trading signal. The smoothed counterpart $P(s_t \mid \mathbf{z}_{1:T})$, which would condition on the full sample, is not used.

The posterior hyperparameters are:

$$\begin{aligned}\kappa_n &= \kappa_0 + n_k, & \mathbf{m}_n &= \frac{\kappa_0 \mathbf{m}_0 + n_k \bar{\mathbf{x}}_k}{\kappa_n}, & \nu_n &= \nu_0 + n_k, \\ \Psi_n &= \Psi_0 + \mathbf{S}_k + \frac{\kappa_0 n_k}{\kappa_n} (\bar{\mathbf{x}}_k - \mathbf{m}_0)(\bar{\mathbf{x}}_k - \mathbf{m}_0)^\top.\end{aligned}\tag{9}$$

New regime parameters are then drawn from the updated posterior: $\Sigma_k \sim \mathcal{IW}(\nu_n, \Psi_n)$ and $\mu_k \mid \Sigma_k \sim \mathcal{N}(\mathbf{m}_n, \Sigma_k / \kappa_n)$.

Step 3: Updating the transition matrix

The sampled path is scanned to count transitions, $n_{ij} = \sum_{t=2}^T \mathbf{1}(s_{t-1} = i, s_t = j)$, the number of times the path moved from state i to state j . Each row of the transition matrix is then drawn from its own Dirichlet posterior:

$$\mathbf{P}_{i,\cdot} \mid \{s_t\} \sim \text{Dir}(\alpha_{i0} + n_{i0}, \alpha_{i1} + n_{i1}).\tag{10}$$

Sampling once from each of the two Dirichlet posteriors (one per regime) produces the updated transition matrix $\mathbf{P}$, which is held fixed during the next iteration’s state-path sampling. Because the prior is strongly tilted toward persistence, the sampled $\mathbf{P}$ tracks the data while staying anchored around a persistent diagonal even on short or noisy paths.

With all three blocks updated, the sampler returns to Step 1. After sufficient iterations the draws converge to samples from the joint posterior.

3.1.5 Implementation and Seed Averaging

The Gibbs sampler is estimated on the training sample only. To reduce sensitivity to MCMC initialization, the full sampler is run independently with $N_s = 200$ different random seeds. Each chain is run for $M = 2,000$ iterations with the initialization described above. The first $B = 500$ iterations are discarded as burn-in, leaving 1,500 posterior draws per seed.

For each seed, posterior mean parameters $\bar{\mu}_k$, $\bar{\Sigma}_k$, and $\bar{\mathbf{P}}$ are computed by averaging over the post-burn-in draws. The forward filter is then applied to the full sample using these parameters to obtain a seed-specific filtered probability $\pi_{t,s}^{\text{filter}}$. The final trading signal is the average across seeds:

$$\pi_t^{\text{filter}} = \frac{1}{N_s} \sum_{s=1}^{N_s} \pi_{t,s}^{\text{filter}}.\tag{11}$$

Averaging across MCMC chains is a methodological requirement, not a performance optimization. A single Gibbs chain produces a noisy regime signal whose downstream Sharpe varies meaningfully across random initializations; averaging across chains reduces this

noise, and the variance shrinks as more chains are added. At 200 chains the Sharpe has effectively stabilized; adding more chains changes it negligibly (Appendix J.4). All out-of-sample numbers reported below are computed from this 200-chain ensemble, not from a single chain.

3.1.6 Regime Identification and Trading Signal

Label Switching and Regime Naming

In a two-state HMM, the labels 0 and 1 are intrinsically arbitrary. Relabeling the states leaves the likelihood unchanged, so the HMM identifies two clusters in the D -dimensional feature space without knowing which one should be called panic. To break this symmetry, each feature is assigned an expected sign during stress (the direction in which it moves during historical crisis windows relative to calm periods), and the cluster whose posterior mean vector best aligns with that pattern is designated as “panic” and the other as “calm”. The specific windows and sign assignments are listed in Appendix I. The HMM itself never sees these windows during training; they enter only at the post-estimation naming step.

The naming has no effect on the regime estimation itself: the two clusters and their boundaries are identical regardless of which is called calm or panic. What the naming controls is the interpretation of the subsequent analysis, which state we treat as the stress regime when reading off mechanism descriptions, conditional Sharpes, and cluster characterizations downstream. The label “panic” therefore reflects a statistical resemblance to historical crisis periods rather than a directly observed market state.

Filtered Regime Probability

Using the posterior mean parameters $\bar{\boldsymbol{\mu}}_k$, $\bar{\boldsymbol{\Sigma}}_k$, and $\bar{\mathbf{P}}$ from Section 3.1.5, the forward recursion (Equation 7) is applied to the full sample to obtain:

$$\pi_t^{\text{filter}} = P(s_t = \text{panic} \mid \mathbf{z}_{1:t}; \bar{\boldsymbol{\mu}}, \bar{\boldsymbol{\Sigma}}, \bar{\mathbf{P}}). \quad (12)$$

A value near 0 indicates high confidence in calm, near 1 indicates panic, and intermediate values reflect genuine ambiguity. This filtered probability serves as the regime signal passed to the portfolio construction methods in Section 3.2.1.

3.2 Stage 2: Regime-Conditioned Portfolio Construction

The filtered regime probability π_t^{filter} from Stage 1 provides a monthly measure of market stress, but on its own it does not select stocks. This stage translates the regime signal into cross-sectional portfolios. Three cross-sectional stock selection methods of increasing complexity are developed: a deterministic regime-adaptive momentum formula (DET), a logistic regression classifier (LR), and an XGBoost gradient-boosted tree regressor (XGB). Each method combines π_t^{filter} with momentum signals at multiple horizons to produce a

monthly score for every stock, which is then used to rank stocks into a long-short portfolio: the top decile (using NYSE breakpoints) forms the long leg, the bottom decile forms the short leg, and both legs are value-weighted. The stock-level inputs (momentum horizons and the regime signal) are described in Section 4.

One distinction from standard momentum strategies matters here. In the traditional approach (Jegadeesh and Titman, 1993), stocks are ranked by their raw trailing return at a fixed lookback, so the long leg always holds the highest-momentum stocks and the short leg the lowest. Here, stocks are ranked by their predicted forward return, a learned function of the regime signal and the full momentum term structure, so the long leg need not contain high-momentum stocks at all. In panic, it can hold stocks below the cross-sectional average at every horizon (Section 5.2.2). The deterministic formula (DET) serves as a transparent baseline. The two machine learning methods go further by jointly learning the regime-momentum interaction.

We evaluate every method in long-short construction rather than long-only, for two reasons. First, the long-short portfolio is dollar-neutral by construction (\$1 long the top decile, \$1 short the bottom decile, both value-weighted), so the realized return reflects whether top-decile stocks outperform bottom-decile stocks rather than market direction. Second, long-short is the standard construction in the momentum literature (Jegadeesh and Titman, 1993; Daniel and Moskowitz, 2016; Barroso and Santa-Clara, 2015), which makes our Sharpe ratios and factor alphas directly comparable to existing results. The value-weighted market (long-only, $\beta = 1$) is reported alongside the long-short rows as a passive equity benchmark, but is not directly Sharpe-comparable with the long-short strategies, since it harvests the equity premium whereas the long-short rows earn their return from cross-sectional ranking.

3.2.1 Cross-Sectional Stock Selection Methods

DET: Deterministic Regime-Adaptive Formula

The simplest approach exploits the economic intuition that the optimal momentum lookback horizon varies with market conditions.⁹ The lookback horizon and stock score are:

$$\ell_t = \text{round}(12 - 11 \cdot \pi_t^{\text{filter}}), \quad (13)$$

$$\text{score}_t^s = \prod_{j=1}^{\ell_t} (1 + r_{t-j}^s) - 1. \quad (14)$$

The first equation sets the lookback horizon as a function of the regime. The second compounds the past ℓ_t monthly returns into a stock score. Stocks are then ranked by this

⁹In the spirit of Goulding et al. (2023), simplified to a continuous panic probability and a single regime-dependent horizon.

score. The top decile forms the long leg and the bottom decile the short leg, as described in Appendix C.1.

At the calm extreme ($\pi_t^{\text{filter}} \approx 0$), $\ell_t = 12$: stocks are ranked by their 12-month cumulative return, with the top decile entering the long leg and the bottom the short. This recovers the 12-month cross-sectional momentum signal of Jegadeesh and Titman (1993). At the panic extreme ($\pi_t^{\text{filter}} \approx 1$), $\ell_t = 1$: the ranking shifts to the most recent monthly return, the horizon at which the short-horizon return dynamics documented by Jegadeesh (1990) and Lehmann (1990) are strongest. Between these extremes, the lookback smoothly interpolates.

LR: Logistic Regression

A logistic regression classifier is trained to predict whether a stock’s forward return exceeds the cross-sectional median in each month. Classification is chosen over direct return regression because monthly stock returns are extremely noisy, and predicting the binary outcome “above or below median” is a more stable learning target that still produces a useful ranking of stocks. Continuous return regression remains the dominant target in the cross-sectional ML asset-pricing literature (Gu et al., 2020). Classification is used here as a deliberately simpler, lower-variance alternative for the linear baseline.¹⁰

$$P(y_t^s = 1 \mid \mathbf{x}_t^s) = \sigma(\boldsymbol{\beta}^\top \tilde{\mathbf{x}}_t^s), \quad (15)$$

where $y_t^s = \mathbb{1}(r_{t+1}^s > \text{median}_s\{r_{t+1}^s\})$ is a binary label indicating whether stock s outperforms the cross-sectional median in the following month. The feature vector

$$\tilde{\mathbf{x}}_t^s = (\text{mom}_{1,t}^s, \dots, \text{mom}_{12,t}^s, \pi_t^{\text{filter}})$$

contains the 12 momentum lookbacks and the regime signal, standardized using training-set means and standard deviations. The model is estimated by maximizing the ℓ_2 -regularized log-likelihood:

$$\hat{\boldsymbol{\beta}} = \arg \max_{\boldsymbol{\beta}} \left[\sum_i \left(y_i \log \sigma(\boldsymbol{\beta}^\top \tilde{\mathbf{x}}_i) + (1 - y_i) \log(1 - \sigma(\boldsymbol{\beta}^\top \tilde{\mathbf{x}}_i)) \right) - \frac{1}{2C} \|\boldsymbol{\beta}\|_2^2 \right], \quad (16)$$

with regularization strength $C = 1.0$. The ℓ_2 penalty shrinks coefficients toward zero to guard against overfitting given the large number of correlated momentum features. Once $\hat{\boldsymbol{\beta}}$ is estimated on the training sample, the stock score for any observation is obtained by plugging $\hat{\boldsymbol{\beta}}$ back into the prediction equation:

$$\text{score}_t^s = \sigma(\hat{\boldsymbol{\beta}}^\top \tilde{\mathbf{x}}_t^s) = P(y_t^s = 1 \mid \mathbf{x}_t^s; \hat{\boldsymbol{\beta}}). \quad (17)$$

¹⁰The logistic sigmoid function $\sigma(z) = 1/(1 + e^{-z})$ maps the linear combination $\boldsymbol{\beta}^\top \tilde{\mathbf{x}}$ from the real line to the interval $(0, 1)$, so the output can be interpreted as a probability.

XGB: XGBoost Regressor

An XGBoost gradient-boosted tree regressor (Chen and Guestrin, 2016) is trained to predict the forward monthly return directly. Unlike LR, which classifies stocks as above or below the median, XGBoost predicts the forward return directly. This preserves magnitude information. A stock expected to gain 10% ranks higher than one expected to gain 2%, a distinction that binary classification discards. The feature vector for each stock-month observation comprises the 13 features described in Section 4.4: the 12 momentum lookbacks ($mom_1, \dots, mom_{12}$) and the regime signal (π_t^{filter}). By including all horizons jointly, XGBoost can learn how the full momentum term structure interacts with the regime signal, for example by conditioning its joint use of all 12 horizons on the level of π_t^{filter} in ways that cannot be decomposed into per-horizon slope changes.

XGBoost builds an additive ensemble of M regression trees, where each successive tree fits the residuals of the current ensemble:

$$F(\mathbf{x}) = \sum_{m=1}^M \eta \cdot h_m(\mathbf{x}), \quad (18)$$

where h_m is the m -th decision tree and η is the learning rate that shrinks each tree’s contribution to prevent overfitting. Each tree h_m is grown to minimize a regularized objective combining the squared-error loss on the current residuals with penalty terms on tree complexity (number of leaves and leaf weights). The predicted return $\hat{r}_{t+1}^s = F(\mathbf{x}_t^s)$ serves as the stock score. The model is trained with $M = 500$ trees, a learning rate of $\eta = 0.05$, maximum tree depth of 4, and row and column subsampling rates of 0.8. To reduce sensitivity to random initialization, an ensemble of 50 independent XGBoost models (each with a different random seed) is trained. Their predictions are averaged before ranking stocks into portfolio deciles.

The maximum depth was selected via cross-validation on the training sample (Appendix G.3). Performance rises monotonically up to depth 4 and declines for deeper trees. Depth 4 is the budget required for the $K = 4$ cluster decomposition in Section 5.2.2, namely depth 1 for the regime gate plus depth 3 for within-regime shape reading.

Among the nonlinear architectures flagged in Section 2.4 (gradient-boosted trees, random forests, neural networks), XGBoost is preferred because its sequential residual fitting concentrates splits on whichever feature most reduces error, turning the regime signal into the model’s primary gating variable rather than one feature among thirteen.

With the two-stage framework in place, the next section describes the data, sample construction, and the feature selection procedure used to choose the HMM’s input features.

4 Data

This chapter chooses the inputs for the regime classifier. We choose which four of nine candidate stress indicators the HMM should use, so that the cross-sectional model can learn the signal generated by the regime as accurately as possible. A four-pass procedure narrows the candidates, shrinking the set on a different criterion at each pass rather than on a single in-sample fit.

The stock-level sample spans January 1990 to November 2024 and includes all ordinary common shares listed on the NYSE, AMEX, or NASDAQ with a price above \$1.¹¹ Following [Shumway \(1997\)](#), delisting returns are incorporated to prevent survivorship bias: actual delisting returns are used when available, and -30% is imputed for performance-related delistings with missing returns.¹² The sample is split into a training period (1990–2010) used to estimate all models and a test period (2011–2024, 167 months) used exclusively for out-of-sample evaluation. No model parameters are re-estimated during the test period. Table 1 provides an overview.

	Train (1990–2010)	Test (2011–2024)	Full
Unique stocks	13,363	7,037	16,657
Stock-month observations	1,161,014	557,087	1,718,101
Avg. stocks per month	4,838	3,336	4,221
Market-level months	240	167	407

Table 1: Sample summary. The sample includes all ordinary common shares listed on NYSE, AMEX, or NASDAQ with price above \$1.

4.1 Feature Selection

The regime signal’s ability to capture the shift in momentum’s cross-sectional structure depends critically on which stress indicators the HMM observes. The selection problem has two constraints. First, market drawdown (DD) is anchored as a baseline feature: [Daniel and Moskowitz \(2016\)](#) show that recent market declines empirically drive the momentum crash, making drawdown the most direct observable measure of the conditions in which the mechanism this thesis studies activates. Second, the number of features is capped at four: with 240 training months, each regime has sufficient observations to estimate a 4-dimensional mean vector and covariance matrix (10 free covariance parameters per state), and more features risk estimation instability.

Candidate pool

Nine monthly macrofinancial indicators, all available from 1990 onward, are considered:

¹¹The \$1 floor follows [Gu et al. \(2020\)](#); value-weighting limits small-stock influence on the realized portfolio.

¹²Full data sources and detailed variable definitions are provided in Appendix A.

- *Equity-market stress*: market drawdown (DD), realized volatility (VOL), and log VIX (LVIX).
- *Cross-sectional structure*: cross-sectional return dispersion (DISP), relative market participation (REL_N), advance-decline ratio (ADR), and cross-sectional return skewness (SKEW).
- *Credit and rates*: BAA–AAA credit spread (CS) and the 10-year minus 2-year Treasury term spread (TERM).

All candidates are observable in real time at the end of each month.

Selection procedure

The selection proceeds through four passes of increasing rigour. In the first pass (*quality screen*), all DD-inclusive combinations of up to four features are considered. With 8 remaining candidates, this gives $1 + \binom{8}{1} + \binom{8}{2} + \binom{8}{3} = 1 + 8 + 28 + 56 = 93$ combinations (DD alone, DD paired with one other, DD with two others, DD with three others). Each is estimated with a single Gibbs sampler seed, and combinations are eliminated if the HMM fails basic sanity checks: the two states must separate meaningfully, and the model must correctly classify known crises (the dot-com bust, the GFC, and COVID) as panic while keeping non-crisis periods calm.¹³

In the second pass (*portfolio ranking*), surviving combinations are evaluated through the full cross-sectional pipeline, testing whether a better regime signal translates into better stock selection. Combinations are ranked by out-of-sample Sharpe ratio at the selection seed budget (10 HMM seeds, 20 XGBoost seeds). The top 10 advance.

In the third pass (*definitive validation*), the surviving finalists are re-evaluated at higher seed counts to confirm that the Pass-2 ranking is not an artefact of low-seed MCMC noise.

In the fourth pass (*stability check*), the finalists are tested for robustness across random seed partitions. A combination that achieves a high Sharpe on one lucky seed combination but collapses on another is unreliable. We run 20 independent experiments with different seed draws and classify combinations as stable ($\sigma < 0.10$), moderate ($0.10 \leq \sigma < 0.15$), or unstable ($\sigma \geq 0.15$).

Selected set

Pass 4 selects DD, DISP, REL_N, and CS as the production set. This combination ranks at the top of the candidate pool on Pass-2 Sharpe and is the most consistent across random seed partitions in Pass 4. It also carries at least one feature from each candidate

¹³Specific thresholds: ESS > 50, regime separation significant on at least $\min(2, D)$ features (where D is the size of the candidate combination), and π_t^{filter} classifies the GFC, COVID, and dot-com crashes as panic while remaining low through calm expansions.

category (equity-market stress, cross-sectional structure, credit and rates), spanning all three dimensions of stress in the candidate pool.

4.2 HMM Input Features

DD is the strongest single-feature classifier (Sharpe 0.66 alone vs 0.97 for the full 4-feature set; Table 13, Appendix B), but leave-one-out diagnostics show the four features contribute on different dimensions. DD and REL_N pull most of the Sharpe weight: removing either drops the full-set Sharpe by 0.46 and 0.54 respectively. DISP and CS are marginal on the Sharpe metric (removal drops it by 0.01 each) but extend the candidate-category footprint (cross-sectional structure and credit-and-rates respectively) and modestly tighten cross-partition σ from 0.14 to 0.11 (Table 12). Each of the four captures a distinct dimension of market stress:

1. **DD (Drawdown):** The percentage decline of the cumulative market price index from its rolling 12-month peak:

$$DD_t = \frac{P_t - \max(P_{t-11}, \dots, P_t)}{\max(P_{t-11}, \dots, P_t)}. \quad (19)$$

DD directly measures the depth of ongoing market declines, making it the natural anchor for regime identification.

2. **DISP (Return Dispersion):** The log cross-sectional standard deviation of individual stock returns:

$$DISP_t = \log(\text{std}_s(r_{s,t})). \quad (20)$$

DISP captures cross-sectional *disagreement* among stocks. During stress, individual stocks move in different directions rather than together, causing dispersion to spike.

3. **REL_N (Relative Market Participation):** The log ratio of the number of actively traded stocks to its trailing 12-month average:

$$REL_N_t = \log\left(\frac{N_t}{\frac{1}{12} \sum_{\ell=1}^{12} N_{t-\ell}}\right). \quad (21)$$

REL_N turns negative during crises when stocks are delisted, trading is halted, or acquisitions accelerate. It captures structural market damage that the other features do not measure.

4. **CS (Credit Spread):** The difference between Moody's BAA and AAA corporate bond yields:

$$CS_t = Y_t^{\text{BAA}} - Y_t^{\text{AAA}}. \quad (22)$$

The credit spread provides an independent signal from the corporate bond market, helping the HMM distinguish genuine stress episodes from equity-specific turbulence.

All four features are winsorized at the 1st and 99th percentiles¹⁴ using training-sample statistics, and standardized to zero mean and unit variance.

The four series co-move around stress episodes: DD and REL_N drop while DISP and CS spike during the 2000–2002 dot-com bust, the 2007–2009 Global Financial Crisis, the March 2020 COVID shock, and the 2022 Fed-hiking cycle, with the shaded NBER recession bands aligning with the largest of these joint excursions. This co-movement makes a single hidden-state variable viable in the HMM.

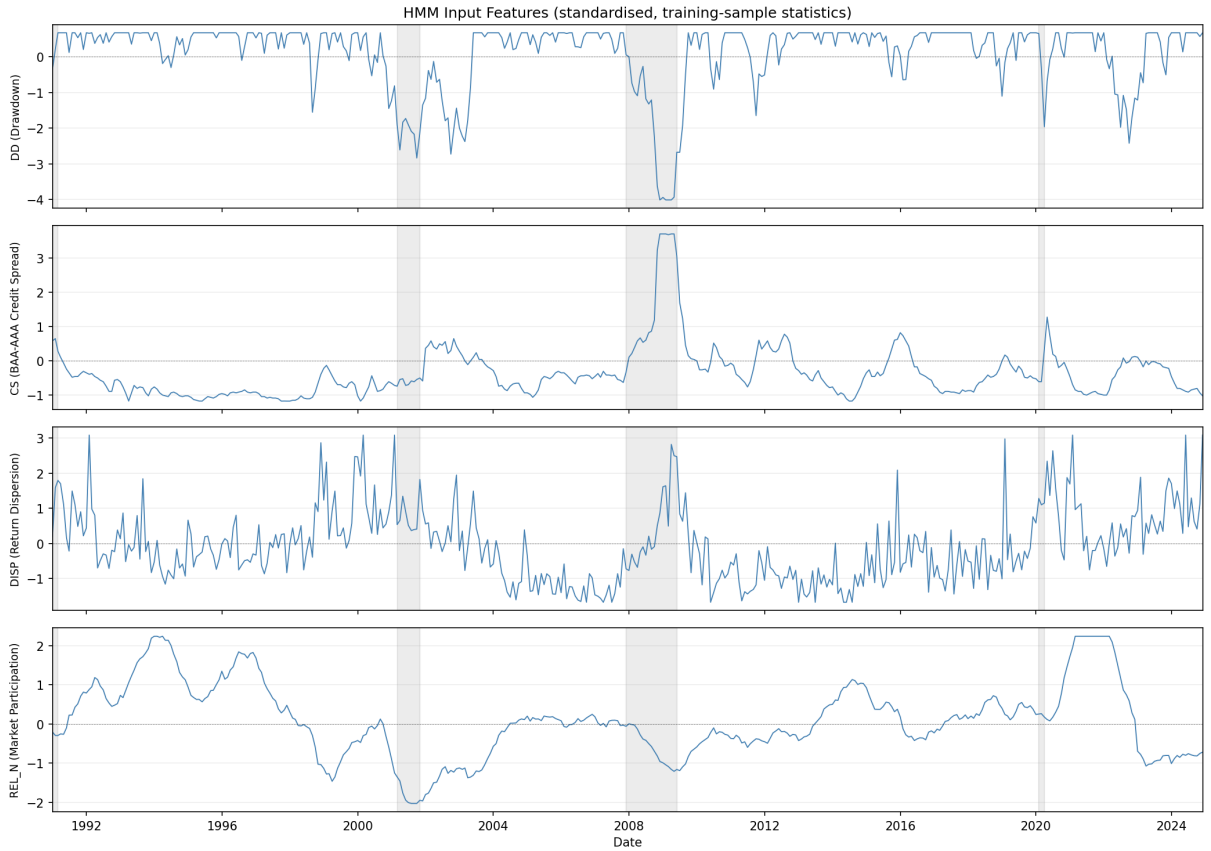


Figure 3: Time series of the four HMM input features (DD, DISP, REL_N, CS), standardized using training-sample statistics. Shaded regions indicate NBER recession periods.

¹⁴Winsorizing at the 1st and 99th percentiles caps any value below the 1st percentile at that bound and any value above the 99th percentile at that bound, limiting the influence of outliers without discarding the observations.

4.3 Regime Estimation Results

Sampler convergence

Convergence of each chain is assessed via the effective sample size (ESS) of post-burn-in draws and the Gelman–Rubin $\hat{R}$ statistic across independent chains, with trace plots as a visual cross-check; formal definitions and the conventional thresholds are collected in Appendix J. On the selected feature set, the Gibbs sampler converges cleanly. All posterior parameters exceed an ESS of 100,¹⁵ so the post-burn-in draws contain at least 100 effectively independent samples per parameter. Across five independent chains drawn from the seed pool, the maximum Gelman–Rubin $\hat{R}$ over all HMM parameters is $\hat{R} = 1.001$ (Table 31, Appendix J.3),¹⁶ well inside the conventional (1.1) and modern (1.01) thresholds.

ESS counts how many effectively independent draws each chain produces. A chain that gets stuck in a small region, or one whose draws are heavily autocorrelated, produces a low ESS. Getting ESS above 100 on every parameter means the posterior has a sharp peak that the chain can navigate, which only happens if the four features genuinely separate calm and panic months. Flat or multi-modal likelihoods cannot reach that threshold, no matter how long the sampler runs.

The Gelman–Rubin $\hat{R}$ checks whether chains started from different initialisations find the same answer. $\hat{R} \approx 1$ means they do; $\hat{R} > 1$ means they do not. With $K = 2$ fixed, the HMM has to produce two clusters. If the data actually had three or more underlying modes, there would be several reasonable ways to compress them into two, and different chains would pick different ones. $\hat{R} = 1.001$ means every chain found the same two-cluster solution, which only happens if the data really is bimodal. Several candidate feature sets were rejected during selection on related convergence grounds, and the clean values here endorse both the four-feature choice and the $K = 2$ specification.

Averaging π_t^{filter} across many independent Gibbs chains stabilizes the regime signal: each chain carries its own seed-specific MCMC noise, and averaging cancels that noise across chains. Table 32 (Appendix J.4) shows the resulting downstream Sharpe rises with the number of averaged chains and plateaus around 30; the 200-chain production setting sits well inside that plateau, so the regime signal is not seed-limited.

Regime separation and filtered probability

Table 2 reports the posterior regime means estimated on the 1990–2010 training period. All four features exhibit statistically significant separation between calm and panic (95%

¹⁵ESS > 100 per parameter is a standard minimum for reliable posterior inference (Gelman et al., 2013); full definition in Appendix J.

¹⁶The Gelman–Rubin $\hat{R}$ compares within- and between-chain variance (Gelman and Rubin, 1992); full definition and thresholds in Appendix J.

credible intervals excluding zero),¹⁷ confirming that the four-feature emission space resolves cleanly into two distinct clusters.

	$\hat{\mu}_{\text{calm}}$	$\hat{\mu}_{\text{panic}}$	Δ	95% CI
DD_z	0.531	-0.976	-1.507	$[-1.794, -1.215]$
DISP_z	-0.423	+0.689	+1.113	$[+0.869, +1.353]$
REL_N_z	0.525	-0.895	-1.419	$[-1.591, -1.231]$
CS_z	-0.697	+0.203	+0.900	$[+0.626, +1.159]$

Table 2: HMM posterior regime-mean separation.

Notes: Posterior mean estimates of regime-conditional feature means from the Bayesian two-state HMM, estimated via Gibbs sampling on the 1990–2010 training period. All features are standardised to zero mean and unit variance. DD = market drawdown; DISP = log cross-sectional return dispersion; REL_N = relative market participation; CS = credit spread (BAA–AAA).

Figure 4 plots the resulting filtered probability π_t^{filter} (constructed as in Section 3.1.6) over the full 1990–2024 sample. The signal spikes at the 2000–2002 dot-com bust, the 2007–2009 GFC, the March 2020 COVID shock, and the 2022 Fed-hiking cycle, and stays near zero during the 2003–2007 and 2013–2019 calm windows. The dashed vertical line marks the train/test split, with the post-2011 portion held out for out-of-sample evaluation.

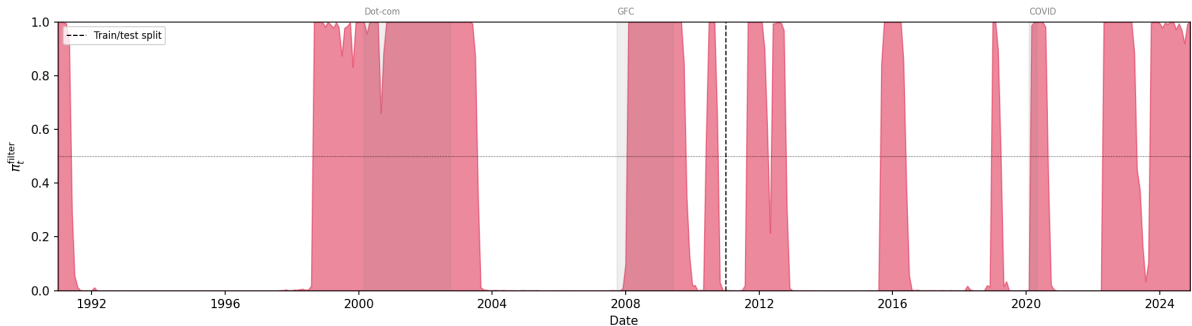


Figure 4: Filtered regime probability π_t^{filter} (1990–2024). The pink area shows the filtered probability path. The dashed vertical line marks the train/test split (January 2011), the dotted horizontal line marks the 0.5 threshold above which a month is classified as panic, and the grey vertical bands mark NBER recession periods. Major stress episodes (Dot-com, GFC, COVID) are labeled at the top.

4.4 Cross-Sectional Features

The cross-sectional models rank stocks each month using twelve trailing momentum signals together with the filtered panic probability π_t^{filter} . The latter varies only at the

¹⁷A 95% credible interval is the Bayesian range covering the parameter with 95% probability; excluding zero means the panic–calm mean difference on each feature is reliably distinct from zero.

monthly market level (every stock in the same month receives the same value), so it functions as a conditioning variable rather than a stock-level predictor.

Twelve trailing momentum signals are computed for each stock, corresponding to lookback horizons of 1 to 12 months:

$$\text{mom}_{k,t}^s = \prod_{\ell=1}^k (1 + r_{t-\ell}^s) - 1, \quad k = 1, \dots, 12, \quad (23)$$

where $r_{t-\ell}^s$ is the adjusted return of stock s in month $t - \ell$. The current month's return is excluded from all lookbacks to avoid mechanical correlation with the forward return being predicted. Including all twelve horizons rather than a single lookback allows the models to learn how the full momentum term structure interacts with the regime signal.

4.5 International Sample

For the robustness check in Section 5.3.4, the same pipeline is applied to UK and Japanese equities from Compustat Global over the same 1990–2024 window. The HMM feature set is re-selected per region using the four-pass procedure described in Section 4.1, run separately on each market's data, with the candidate pool adapted to use locally available stress indicators. Both regions select *DD+VOL+REL_N*. Long and short legs are formed from unconditional decile breakpoints across all listed stocks, since no NYSE-equivalent tier exists for either market. Per-region selection details and stability checks are in Appendix K.

With the regime signal and the cross-sectional features in place, the next chapter evaluates the three portfolio-construction methods out-of-sample and opens up the chosen model's selections.

5 Results

This chapter answers how the cross-sectional structure of momentum changes across regimes. Of the three portfolio-construction methods, only the XGBoost variant captures the regime-momentum interaction; opening it up month by month reveals four recurring selection modes running from classical winner selection in calm regimes to full term-structure reversal at panic peaks. The result survives a risk-aversion sweep, a 25-year expanding-window backtest covering the dot-com bust and the GFC, and an international replication on UK and Japanese equities. All strategies are evaluated on the 167-month 2011–2024 long-short out-of-sample window.

5.1 Portfolio Performance

5.1.1 Results Overview

	Ann. Ret	Ann. Vol	Sharpe	Max DD	β	NW t	Final \$
Market	11.9%	14.6%	0.84	-24.7%	1.00	–	4.8
Fixed 12-mo mom	-1.9%	26.1%	0.06	-72.2%	-0.66	0.24	0.8
Fixed 1-mo mom	3.1%	17.6%	0.26	-30.2%	-0.34	1.08	1.5
DET	-0.2%	23.6%	0.11	-63.8%	-0.59	0.43	1.0
LR	-1.3%	15.2%	-0.01	-42.4%	0.05	-0.04	0.8
XGB	21.7%	19.5%	1.11	-22.8%	0.45	4.37***	15.3

Table 3: Out-of-sample portfolio performance.

Notes: Long-short (top/bottom decile, NYSE breakpoints), value-weighted, net of 10 bps one-way costs. β is the market beta. NW t uses Newey–West standard errors (6 lags). Test period: January 2011 to November 2024 (167 months). * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 3 reports out-of-sample performance¹⁸ for the three cross-sectional model variants (DET, LR, XGB) alongside three benchmarks: the value-weighted U.S. market and two unconditional momentum portfolios at the dominant 12-month and contrarian 1-month horizons. The model variants share the same inputs, the twelve monthly momentum horizons together with the HMM filtered probability π_t^{filter} , but differ in functional form: DET is a deterministic rule that interpolates between mom_1 and mom_{12} as π_t^{filter} varies; LR is a logistic regression on those features; XGB is an XGBoost regressor capable of representing arbitrary nonlinear interactions.

The pattern across rows is clear. The value-weighted market earns 11.9% annualised

¹⁸Sharpe convention details (raw vs. excess returns, arithmetic-vs-geometric mean) are in Appendix C.3.

at Sharpe 0.84, reported as the passive equity benchmark.¹⁹ Unconditional 12-month momentum (the canonical anomaly) delivers only Sharpe 0.06 with a -72.2% maximum drawdown, exposing the crash vulnerability documented by Daniel and Moskowitz (2016) and Barroso and Santa-Clara (2015). The contrarian 1-month version reaches 0.26. Of the regime-conditional variants, the deterministic rule (DET) improves on 12-month momentum only marginally to 0.11, and the logistic regression (LR), given identical features to XGB, collapses to -0.01 . Only XGB delivers economically meaningful performance, an annualised Sharpe of 1.11 with a six-factor alpha of 24.1% ($t = 4.81$; Appendix H.3).²⁰ It is the only row in Table 3 whose Newey-West t -statistic clears every conventional threshold. Block-bootstrap inference confirms this gap is not driven by sampling noise (Appendix H.4).

A further specification confirms that the limiting factor is linearity itself. Hand-engineering the missing nonlinearity into LR through π^2 and $\pi \times mom_k$ terms lifts it to 0.32, establishing that the regime-momentum interaction is real while also showing that pre-specified interactions recover less than a third of what XGBoost finds automatically. Adding firm-level fundamentals to XGB’s feature set degrades performance rather than improving it (Section 5.1.3), evidence that π_t^{filter} ’s role is specifically to condition the momentum term structure. XGB is therefore adopted as the production model for the remainder of the thesis. Figure 5 traces the cumulative wealth paths of each row from January 2011; XGB’s advantage concentrates in the shaded panic months, motivating the regime-by-regime decomposition that follows.

¹⁹The market is long-only with full market exposure ($\beta = 1$), while every other row in Table 3 is long-short and approximately market-neutral, so raw Sharpe ratios are not directly comparable across the long-only/long-short divide: the market harvests the equity premium, whereas the long-short rows earn their return from cross-sectional ranking. The like-for-like Sharpe peers for the model variants are therefore the unconditional long-short momentum benchmarks.

²⁰The factor alpha is the intercept of a regression of long-short returns on the market, SMB, HML, RMW, CMA, and UMD factors. Including UMD is the toughest version of the test for a momentum strategy: the regression cancels any return earned by mechanically tilting toward winners. Full specification in Appendix H.3.

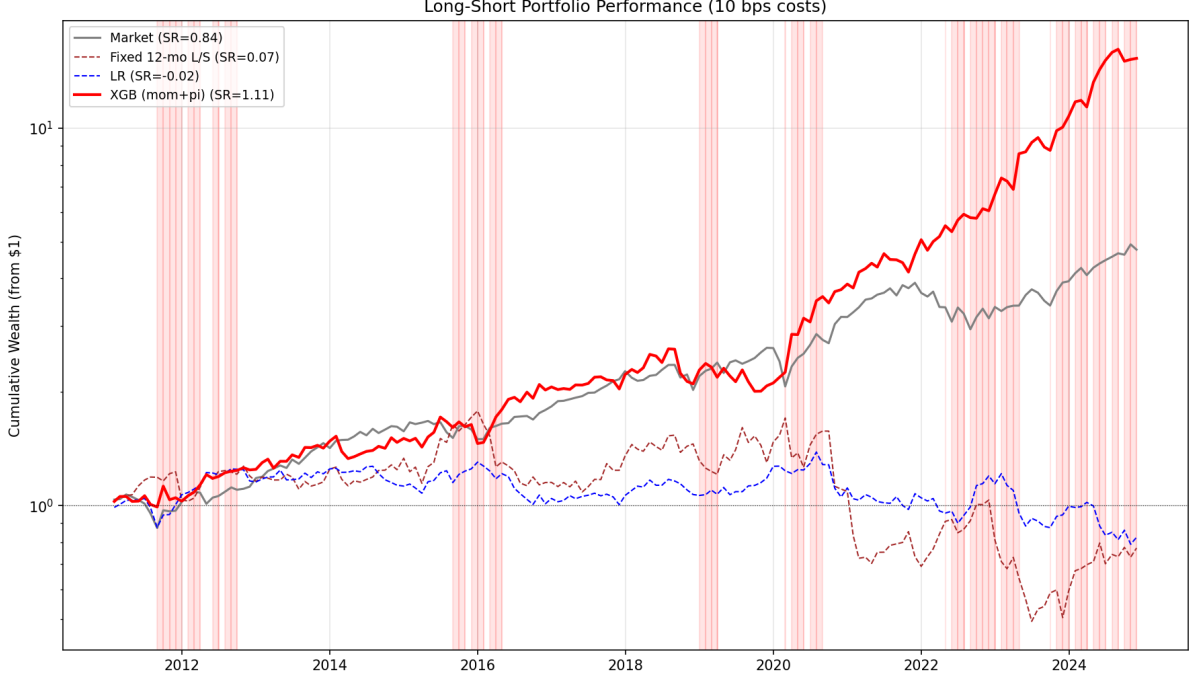


Figure 5: Cumulative wealth paths (log scale) starting from \$1 in January 2011. Shaded bands indicate panic months ($\pi_t^{\text{filter}} \geq 0.5$).

The regime-conditional view (Table 4) shows where this advantage comes from: XGB’s panic Sharpe (1.53) is roughly double its calm Sharpe (0.84), exactly the regime-conditioning pattern the architecture is designed to produce, and exactly where unconditional momentum is known to crash.²¹

	Full	Calm	Panic
Market	0.84	0.64	1.13
Fixed 12-mo mom	0.06	0.08	0.05
Fixed 1-mo mom	0.26	0.52	-0.01
DET	0.11	0.18	-0.01
LR	-0.01	-0.29	0.39
XGB	1.11	0.84	1.53

Table 4: Regime-conditional Sharpe ratios (108 calm months, 59 panic months).

5.1.2 Regime Signal Ablation

The regime signal is essential. Removing π_t^{filter} and retraining on the 12 momentum features alone drops the Sharpe to 0.43 (Table 5). Replacing π_t^{filter} with the four raw

²¹The point-estimate gap is not statistically significant at the 5% level on the 59 available panic months (block-bootstrap paired test, $p = 0.109$; Appendix H.4); confirmation would require a longer panic sample. The same table shows every benchmark trailing XGB by a wide margin in panic (LR: 0.39, Fixed 12-mo: 0.05, against XGB’s 1.53).

stress indicators (DD, DISP, REL_N, CS) performs even worse (Sharpe 0.26). With four additional dimensions the model must reconstruct the regime structure at each split, while the HMM distils those indicators into a single dimension that one tree split can condition on. The HMM’s role is not to provide a convenient feature but to solve regime identification *before* stock selection, freeing the cross-sectional model to use the regime signal rather than construct it.

Regime Signal Variant	Ann. Ret	Ann. Vol	Sharpe	Max DD
XGB + π_t^{filter} (HMM)	21.7%	19.5%	1.107	−22.8%
XGB + raw indicators (DD, DISP, REL_N, CS)	3.2%	19.1%	0.257	−45.4%
XGB (no regime signal)	7.0%	20.4%	0.429	−41.0%

Table 5: Regime signal ablation (50-seed ensemble). The HMM signal outperforms both raw stress indicators and no regime signal.

5.1.3 Fundamental Feature Augmentation

A complementary question is whether *adding* widely used firm-level fundamentals to the production feature set strengthens the strategy. Ten characteristics are tested,²² each trained at production settings.

	Feat.	Sharpe	Calm	Panic	Fund SHAP
Baseline (mom + π)	13	1.11	0.84	1.53	0%
Baseline + 10 fundamentals	23	0.89	0.69	1.18	44%
Fundamentals only	10	0.59	0.86	0.38	100%
Fundamentals + π	11	0.64	0.59	0.72	60%
Momentum + fund. (no π)	22	0.63	0.68	0.57	54%

Table 6: Feature-set ablation (50-seed XGBoost ensembles, production hyperparameters). Full Sharpe is reported alongside calm-period and panic-period Sharpe. “Fund SHAP” is the share of total SHAP attention absorbed by the ten fundamental features.

Adding the ten fundamentals on top of the production feature set reduces Sharpe from 1.11 to 0.89 (Table 6), with the loss concentrated in panic months: panic Sharpe falls from 1.53 to 1.18 while calm Sharpe drops only from 0.84 to 0.69. The fundamentals absorb 44% of total SHAP, redirecting attention from the momentum term structure (54% to 27%) and the regime signal (46% to 29%).

Fundamentals describe firm-level characteristics whose cross-sectional ranking does not flip with the market regime the way the optimal momentum tilt does. A high-B/M stock is a high-B/M stock in both regimes, so fundamentals cannot reproduce the panic-side reversal that drives the baseline’s panic Sharpe (Table 6, fund-only and fund + π

²²Book-to-market, return on equity, gross profitability, asset growth, leverage, earnings growth, log market equity, operating cash flow over assets, free cash flow over assets, and accruals. All real-time and four-month-lagged from Compustat `fundq`, with full definitions in Appendix L.

rows). Adding them on top of the regime-momentum signal therefore dilutes an essential mechanism rather than complementing it. A regime-conditional feature-set design that routes fundamentals into calm-regime predictions and momentum into panic-regime predictions is discussed in Section 6.3.

5.2 The Regime-Momentum Mechanism

This section opens the XGBoost black box, reading the structure of the regime-momentum signal directly off the model’s per-month selections.

We first analyze the contribution by horizon (Section 5.2.1), then dig deeper into the common decision patterns the model produces (Section 5.2.2), and finally discuss the results (Section 5.2.3).

5.2.1 Momentum Contribution by Horizon

XGB’s predictions are decomposed using SHapley Additive exPlanations (SHAP) values (Lundberg and Lee, 2017), which measure how much each input contributes to the model’s prediction (definitions in Appendix D.1). A single feature, π_t^{filter} , carries 46% of total SHAP, with the 12 momentum horizons together accounting for the other 54% (Table 7). That 46% from one of thirteen features is striking: equal weighting would give π_t^{filter} just $1/13 \approx 8\%$, so the model treats the regime signal as roughly six times its share of the feature set. The concentration on a single feature supports a routing-variable role: π_t^{filter} determines how the model uses the momentum term structure. Between regimes, momentum’s share rises from 51% in calm to 59% in panic while π_t^{filter} ’s share falls from 49% to 41%. The regime signal does most of its work where the regime call is ambiguous (calm and the calm/panic boundary) and recedes once π_t^{filter} saturates near 1 in deep panic.

Feature	Overall	Calm	Panic
Momentum (12 horizons)	54%	51%	59%
π_t^{filter}	46%	49%	41%

Table 7: SHAP feature importance shares for XGB. Each column sums to 100%.

The aggregate split is reported by leg and horizon in Figure 6. Both legs concentrate on months 8–12 (about 65% each), with months 11 and 12 carrying the largest individual shares. The short leg’s peak at month 11 (22%) is the most concentrated single-horizon share in the figure. Both legs’ concentration on the 8–12 month window is consistent with the classical Jegadeesh and Titman (1993) 12-month momentum range: this is where the cross-section’s momentum signal is historically strongest, whether the model is identifying winners or losers. The short leg’s tighter concentration at month 11 (versus the long leg’s

flatter spread across 8–12) suggests loser identification is more horizon-specific than winner identification.

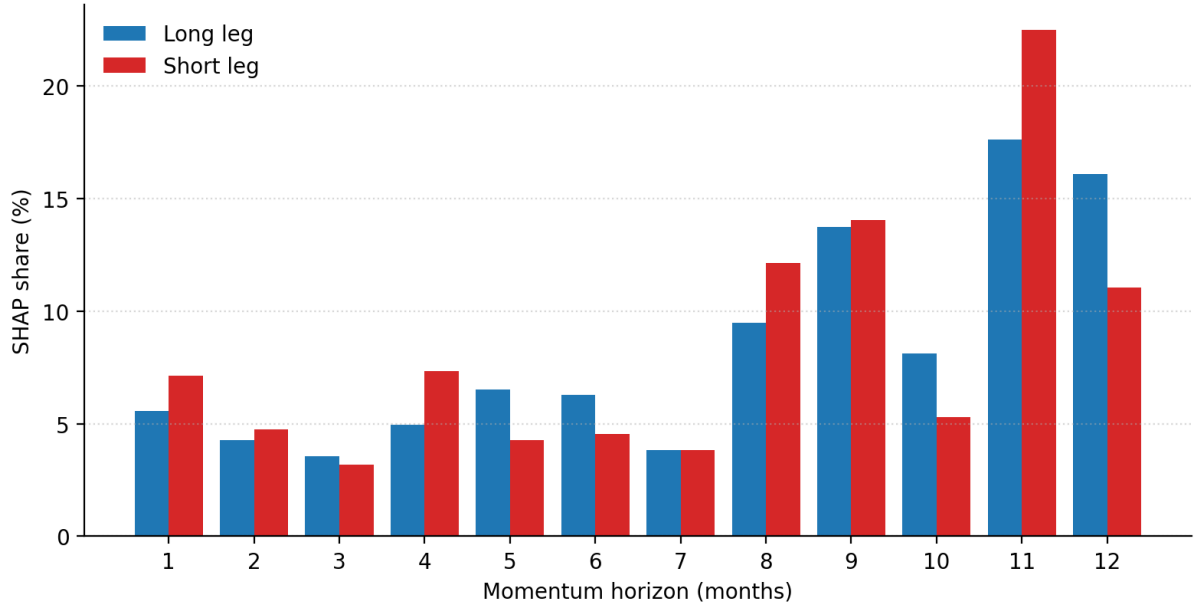


Figure 6: Per-horizon SHAP shares of momentum features in XGBoost’s predictions, by leg, pooled across calm and panic regimes weighted by stock-month counts.

SHAP shares show which horizons matter, but not in which direction: the same SHAP weight could reflect winner-selection or loser-selection at that horizon. Whether XGB is treating the momentum term structure as a continuation signal, a reversal signal, or something that flips with the regime is a question the aggregate SHAP table cannot answer. Section 5.2.2 resolves this by tracing the picks themselves through their cross-sectional z-scores.

5.2.2 Cross-Sectional Z-Scores

To trace the picks themselves, we look at the cross-sectional z-scores of the stocks XGB selects for the long leg. For each stock s the model has placed in the long leg, the z-score at horizon k is $(mom_k^s - \bar{mom}_k) / \sigma_k$, where $\bar{mom}_k$ and σ_k are the cross-sectional mean and standard deviation of horizon- k trailing momentum across all stocks in the universe that month. The long-leg z-curve for a given month is the average of these per-stock z-scores across all stocks the model selected for the long leg, giving a 12-element profile describing where the model’s picks sit relative to the cross-sectional mean at each lookback. The sign tells us which side of the cross-sectional mean the basket sits on at horizon k (positive: above the mean; negative: below). The magnitude tells us how far it sits in standard-deviation units: a z-score of +0.8 means the model’s picks have, on average, trailing momentum at horizon k that is 0.8 cross-sectional SDs above the average that month, a more pronounced tilt than +0.2. Figure 7 plots each month’s long-leg z-curve as one row.

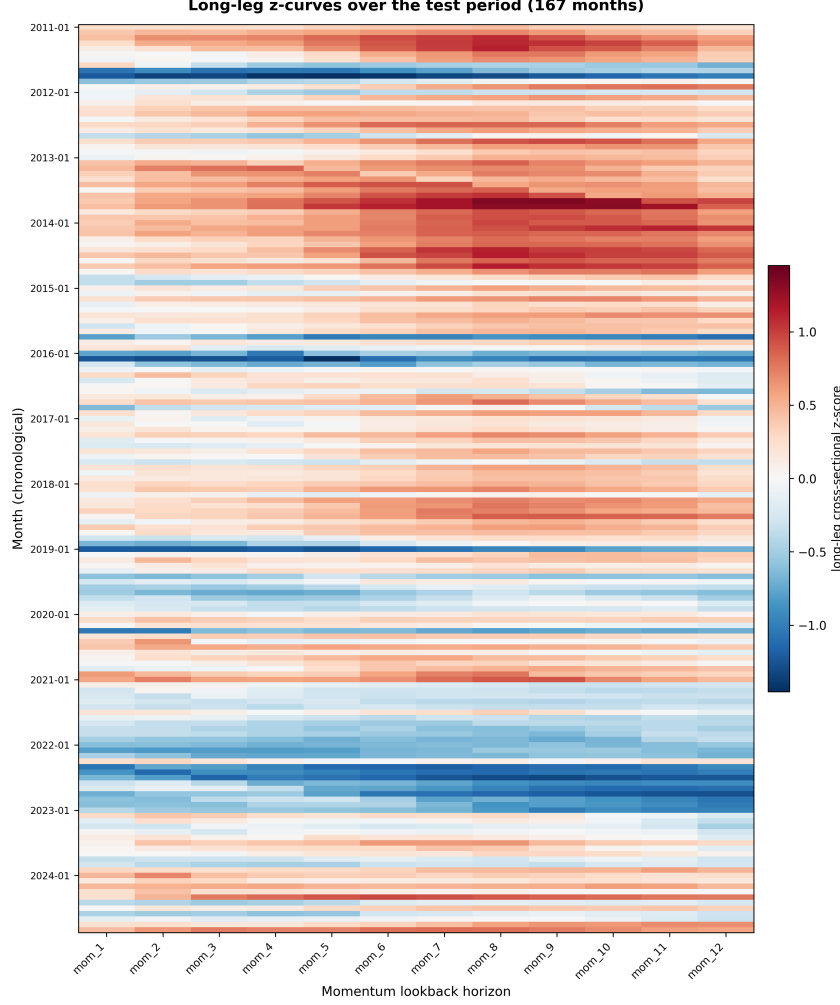


Figure 7: Long-leg cross-sectional z-scores by month and momentum horizon, 2011–2024. Each row is one of 167 test months sorted chronologically. Each column is one of the 12 momentum horizons. Color shows the deviation of the long leg’s mean momentum at horizon h from the cross-sectional mean trailing momentum that month.

The heatmap reveals a small number of recurring selection modes. To summarize these, we cluster the 167 monthly z-curves with K -means using Euclidean (L_2) distance. Each month’s z-curve is a point in 12-dimensional space, one coordinate per momentum horizon. The L_2 distance between two months A and B is the square root of the sum of squared per-horizon z-score differences, $\sqrt{\sum_{k=1}^{12} (z_k^A - z_k^B)^2}$, so two months are close when their long-leg baskets sit at similar z-scores at each of the twelve horizons. K -means partitions the 167 months into $K = 4$ groups by minimizing total within-cluster L_2 distance to each group’s centroid. $K = 4$ is chosen on economic grounds. We do not use $K = 3$ because the coarser partition collapses the two below-average reversal modes (post-panic recovery and deep-crisis term-structure-wide reversal) into a single cluster, eliminating a distinction central to the mechanism (Section 5.2.3). We do not use $K \geq 5$ because the four-cluster partition already covers the two above-average modes and two below-average modes that the mechanism turns on, and further splitting would partition

an economically homogeneous group without revealing a new selection pattern.

The partition is stable across random initializations: 97.6% of months get the same label across six seeds, with mean pairwise Adjusted Rand Index 0.96. The four groups nest inside a coarser two-fold partition. Clusters 1 and 2 are the 115 months in which the long leg sits above the cross-sectional mean (“above-average baskets”, $\bar{\pi} = 0.28$). Clusters 3 and 4 are the 52 months below (“below-average baskets”, $\bar{\pi} = 0.52$).

Per-cluster descriptive statistics are in Table 8. The full feature signature is in Appendix Table 26. Each cluster’s z-curve panel appears immediately above its description (Figures 8–11). Thin lines are member-month z-curves and the bold line is the cluster centroid.

Two terms recur in the cluster descriptions below. The *cross-section* is the full universe of stocks taken into consideration at a given month, and *cross-sectional* statistics are computed across this universe (the cross-sectional mean trailing return at horizon k , for example, is the average horizon- k trailing return taken over every stock in the universe that month). Each cluster description then reports two return profiles side by side: the *cross-sectional mean trailing return* (averaged across the universe, describing the market state in that cluster) and the *long-leg mean trailing return* (averaged across the stocks XGB picked, describing where the model positioned within that market). The first tells us what the market did; the second tells us how the picks compare to it. Two further cross-sectional return statistics also recur: *dispersion* (the spread of returns across stocks at that horizon, i.e., how scattered they are) and *skewness* (the asymmetry of the return distribution, positive when a few outlier winners stretch the right tail, near zero when stocks move together symmetrically).

Feature	1 (calm)	2 (transition)	3 (post-panic)	4 (deep crisis)
Months (n)	53	62	31	21
$\bar{\pi}_t^{\text{filter}}$ (current panic prob.)	0.21	0.34	0.32	0.81
Cross-section std of 9–12 mo momentum	0.60	0.59	0.85	0.45
Cross-section skewness of 1–4 mo momentum	4.6	7.2	6.4	3.7
Long-leg avg z-score $\bar{z}$	+0.57	+0.19	−0.29	−0.82
Long-leg avg trailing return	+40%	+16%	−3%	−37%
Cross-section avg trailing return	+14%	+7%	+13%	−9%
Short-leg avg trailing return	−3%	+0%	+42%	−4%
Cluster Sharpe	0.92	0.93	1.21	1.82

Table 8: $K = 4$ cluster descriptors, 2011–2024 ($n = 167$ months); columns are clusters. The trailing-return rows (long, cross-section, short) are averaged across the 12 momentum horizons. Dispersion is shown at 9–12 months and skewness at 1–4 months, the windows where clusters differ most. Cluster Sharpe is annualised, block-bootstrapped (block size 6, 5,000 reps).

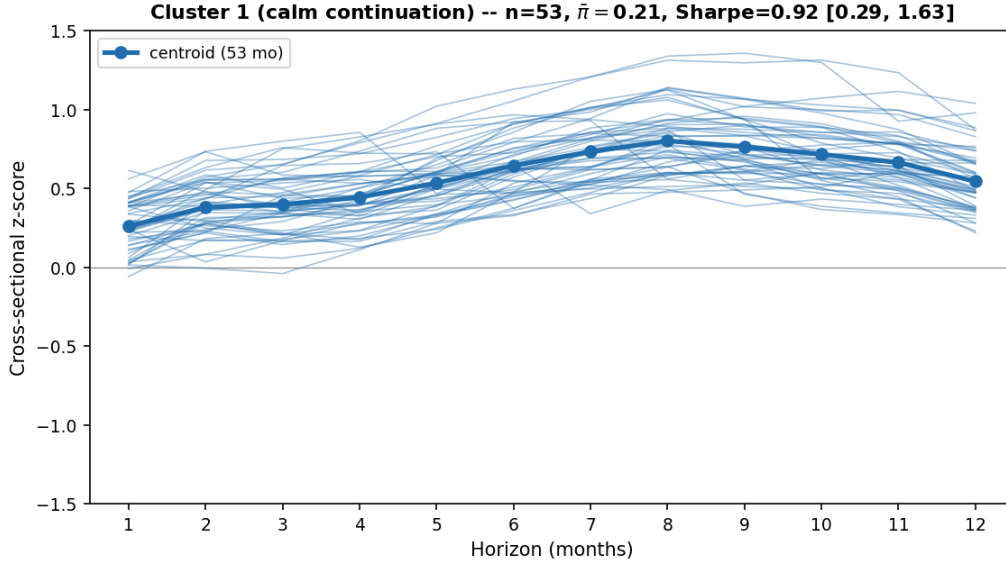


Figure 8: Cluster 1 (calm continuation): z-curves of the 53 member months and centroid. The centroid is humped and uniformly positive, peaking at +0.80 at month 8.

Cluster 1 (calm continuation): a quiet, broadly trending market

The long-leg z-curve has a z-mean of +0.57, the largest average amplitude on the continuation side, with a humped shape. Picks sit only mildly above average at short horizons ($z \approx +0.26$ at month 1) but are well differentiated at intermediate horizons (peak +0.80 at month 8). In absolute terms, the long-leg mean trailing return reaches +60% at the 9-month horizon. The cluster is concentrated in 2013–2014, running in stretches of up to 21 consecutive months.

These 53 months are a stable, mature bull market. With cluster-mean $\bar{\pi}_t^{\text{filter}} = 0.21$ (the lowest of any cluster), the model has high confidence the regime is calm and commits fully to its standard winner-selection mode. The cross-sectional mean trailing return is positive at every horizon (the average stock returned +3%/+13%/+23% over 1/6/12 months), so there is a coherent up-trend for the model to exploit. The rise is broad rather than driven by a few outliers (cross-sectional return skewness 4.6/6.2/6.7, below the test-period average of 5.8/6.6/7.1), which means winner-selection picks up genuine widespread participation rather than chasing a handful of unrepresentative names. The cluster sits far from the post-bear-market rebounds where unconditional momentum is known to crash.

This is the canonical setting for cross-sectional momentum, the kind of extended bull market in which [Cooper et al. \(2004\)](#) document that momentum profits are most concentrated. The picks reproduce the buy-side of [Jegadeesh and Titman \(1993\)](#)’s twelve-month relative-strength strategy, with the intermediate-horizon peak matching [Novy-Marx \(2012\)](#)’s documented dominance. The hump’s low end at month 1 is consistent with the low SHAP share month 1 receives in [Section 5.2.1](#), and with the short-horizon

reversal and microstructure noise long documented in the literature (Jegadeesh, 1990; Lehmann, 1990). The underlying mechanism is Hong and Stein (1999)’s gradual information diffusion, in which news enters prices slowly so winners keep winning until the information is fully absorbed.

Cluster 1 earns a Sharpe of 0.92, the lowest of the four.²³ Continuation still pays, but the strategy is following an established rally rather than catching a turning point. Cluster 1 serves as a sanity check on the architecture: at the lowest π_t^{filter} values in the test sample the model recovers the classical cross-sectional momentum pattern, so the regime-conditioning architecture adds new behavior during stress without overriding the unconditional pattern that already works in calm.

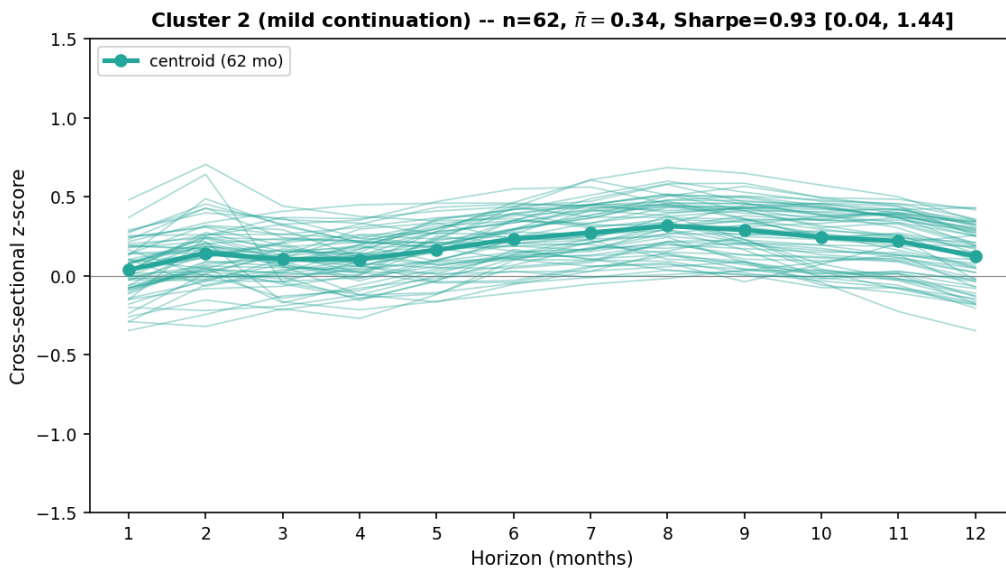


Figure 9: Cluster 2 (transition): z-curves of the 62 member months and centroid. The centroid is mildly positive throughout, peaking at +0.32 at month 8.

Cluster 2 (transition): a choppy market with no broad direction

The long-leg z-curve is nearly flat, with most horizons sitting between +0.10 and +0.30 (z-mean +0.19, peak +0.32 at month 8), the smallest absolute amplitude of any cluster, reflecting a hedged selection rather than a confident commitment. Cluster 2 is the largest of the four ($n = 62$) but appears only in short bursts (max 4 consecutive months) spread throughout the test period.

These 62 months are an ambiguous transition state, with cluster-mean $\bar{\pi}_t^{\text{filter}}$ in a middle band (0.34, neither clearly calm nor clearly panic). The cross-sectional average momentum is mild ($\text{mom}_{\text{overall}} = 0.07$, below the test-period average of 0.08), so no clear up-trend has formed. The cross-sectional mean trailing return is only mildly positive (the average stock

²³Cluster Sharpe is computed from the realized long-short monthly returns over the months assigned to that cluster: $\text{Sharpe}_C = \text{mean}(r_{M_C}) / \text{std}(r_{M_C}) \cdot \sqrt{12}$, where M_C is the set of months in cluster C . Confidence intervals come from a block bootstrap (block size 6, 5,000 reps).

returned +2%/+8%/+11% over 1/6/12 months) and cross-sectional return skewness is the highest of any cluster (7.2/7.6/8.1), with a small number of extreme winners pulling the mean above zero while the median stock's return stays near flat, so even the modest positive level is not broad-based. With π_t^{filter} between calm and panic without decisive separation, the model plays safe and earns its return through occasional outlier capture rather than systematic selection. The short-burst pattern is consistent with cluster 2's role as a brief transition between regimes rather than a stable state. The muted shape is consistent with [Cooper et al. \(2004\)](#)'s finding that momentum profits concentrate in clearly bullish prior states, fading when the market state is ambiguous, and with [Hong et al. \(2000\)](#)'s slow-diffusion mechanism, in which momentum profits concentrate in stocks completing their delayed information adjustment rather than spreading across the cross-section.

Cluster 2 earns a Sharpe of 0.93, essentially identical to cluster 1's 0.92 but reflecting a different mechanism: where cluster 1 exploits a confidently-tilted winner basket, cluster 2 picks up its return through occasional right-tail outliers rather than systematic selection. The cluster exposes the limit of the regime-momentum architecture, since when π_t^{filter} sits in the ambiguous middle and the cross-section of stocks carries no clear directional trend, the model cannot build a confidently-tilted long leg even with the regime signal available.

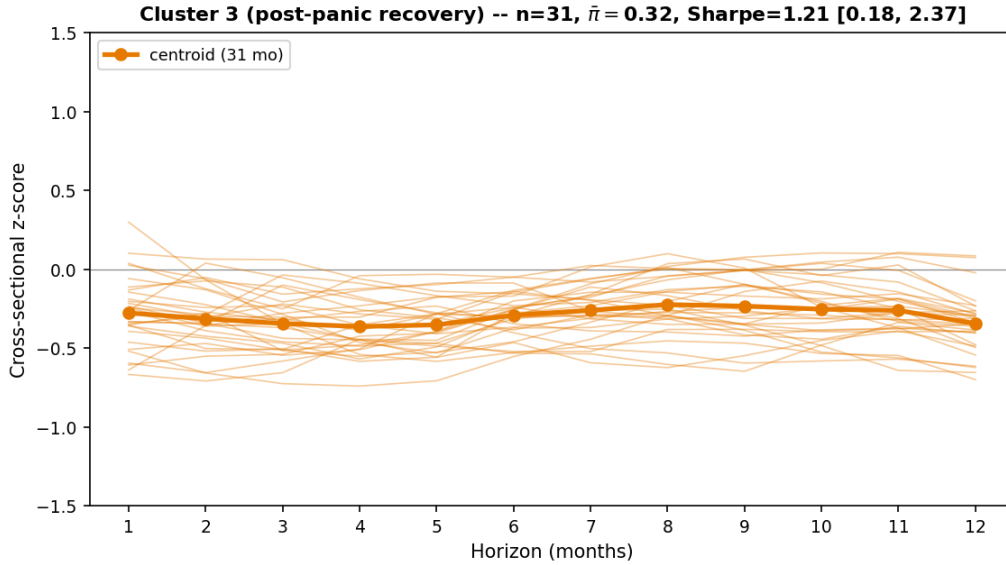


Figure 10: Cluster 3 (post-panic recovery): z-curves of the 31 member months and centroid. The centroid sits below the cross-sectional mean at every horizon, between -0.22 and -0.36 .

Cluster 3 (post-panic recovery): a recovering market with high long-horizon dispersion

The long-leg z-curve sits below the cross-sectional mean at every horizon, with z-mean -0.29 . The deepest below-average points are at month 4 (-0.36) and month 12 (-0.35).

The amplitude is moderate, smaller than the extreme commitments seen in clusters 1 and 4. The long-leg mean trailing return ranges from -6% to $+1\%$ across horizons, while the cross-sectional mean trailing return is up double digits. The picks rotate more in this cluster than in any other. Only about 68% of the long-leg names carry over from one cluster-3 month to the next consecutive cluster-3 month (the lowest carry-over fraction of any cluster), as some recovery candidates catch up and others remain behind. Cluster 3 is concentrated in 2021 (9 of 31 months, the post-COVID rotation) and 2023.

These 31 months sit at a moderate cluster-mean panic probability ($\bar{\pi}_t^{\text{filter}} = 0.32$) paired with a strongly recovering cross-sectional mean trailing return (the average stock returned $+1\%/+11\%/+27\%$ over 1/6/12 months). The strong 12-month return paired with a near-flat most recent month tells us the rally is mature. The bulk of the rebound has already happened in the prior 6–12 months, and cluster 3 catches the late-stage decelerating phase. The joint signature of moderate π and strong positive cross-sectional average momentum conditions the trees to select from a post-bear-rebound branch, where the long leg is built from cross-sectional losers rather than winners. Cross-sectional return dispersion is the highest of any cluster at both the 5–8 and 9–12 month horizons (0.56 and 0.85), reflecting an uneven recovery in which some stocks rebound while others still lag. The increasing dispersion across horizons matches [Lee and Swaminathan \(2000\)](#)’s lifecycle interpretation, in which momentum signals evolve from continuation to reversal as information ages.

These months also exhibit the beta-reversion dynamic that [Daniel and Moskowitz \(2016\)](#) identify as crash-prone for unconditional momentum. During the preceding bear market, high- β stocks fell disproportionately and low- β defensives held up; when recovery arrives, the high- β names rally disproportionately and the defensives lag. Fixed 12-month momentum therefore enters the recovery on the wrong side of the trade, long the defensives and short the high- β rebounders, and loses money. Conditioning on π_t^{filter} reverses XGB’s picks: it puts the high- β rebounders in its long leg and the defensives in its short leg, so the same rally that crashes unconditional momentum lifts XGB instead. Cluster-level CAPM betas confirm this (Section 5.2.3 below and Appendix H.7, Table 27).

Cluster 3 earns a Sharpe of 1.21, in the same months where classical momentum is most vulnerable. What is distinctive is the counter-intuitive bet that selecting stocks still below average in a recovering market is more profitable than chasing the names that have already recovered.

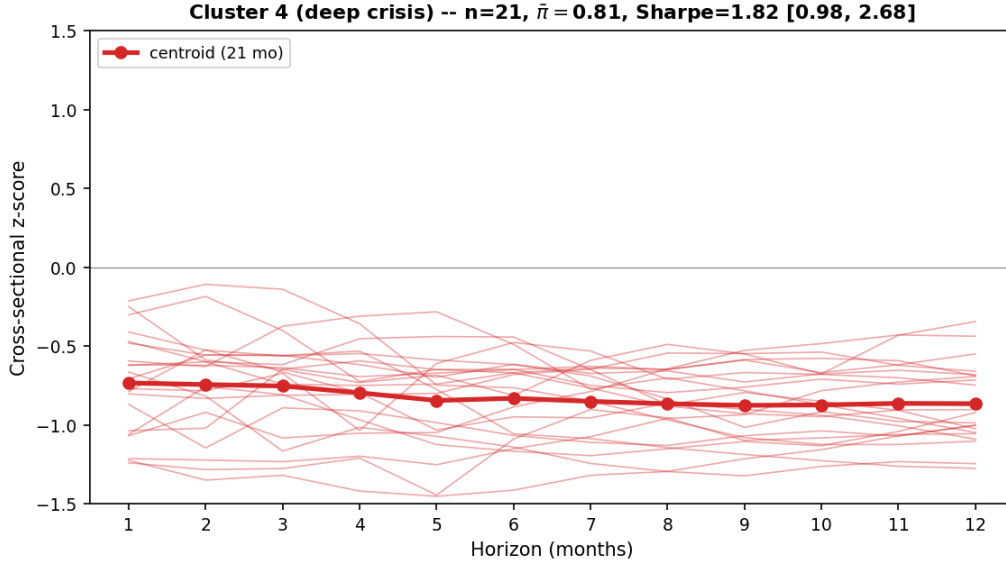


Figure 11: Cluster 4 (deep crisis): z-curves of the 21 member months and centroid. The centroid sits uniformly between -0.73 and -0.88 across all 12 horizons.

Cluster 4 (deep crisis): an active panic with the whole market falling

The long-leg z-curve sits uniformly between -0.73 and -0.88 across all 12 horizons (deepest around month 9, z-mean -0.82), meaning the picks are roughly 0.7 to 0.9 cross-sectional standard deviations below the cross-sectional mean at every horizon. The long-leg mean trailing return deepens from -15% over the prior month to -48% over the prior 12 months. Eleven of 21 months are in 2022 (Fed-hiking cycle), with corroborating instances in April 2020 (COVID trough), October 2015 and January 2016 (China devaluation, oil collapse), and January 2023.

The market is in active panic, with cluster-mean $\bar{\pi}_t^{\text{filter}} = 0.81$ (the highest of any cluster) and the cross-sectional mean trailing return uniformly negative (the average stock returned $-5\%/-11\%/-7\%$ over 1/6/12 months). The 6-month return is more negative than the 12-month return, so the older half of the year was roughly flat or positive and the steep decline is concentrated in the most recent months: the signature of a short panic rather than a prolonged year-long deterioration. Cross-sectional return dispersion is the lowest of any cluster at both the 5–8 and 9–12 month horizons (0.35 and 0.45), and 1–4 month cross-sectional return skewness is also the lowest (3.67), so stocks are moving down together rather than fanning out.

Shocks of this kind (a Fed pivot, a lockdown, a credit event) historically resolve within months, so the model reads the deeply discounted high- β losers as mispriced rather than terminally impaired. With π at this extreme, it commits fully to deep-crisis loser-selection, positioning *for* the same high- β rebound that cluster 3 *harvests* during recovery. Cluster 4 is therefore the anticipatory counterpart to cluster 3's realized one, activating at the panic peak on the premise that the panic resolves quickly. Section 5.3.3 examines what

happens when it does not.

Cluster 4 earns the highest Sharpe in the partition, 1.82, on a mean monthly return of +3.13%. The rebound bet pays off in 13 of the 21 cluster-4 months (62%), with winners comfortably outsizing losers: the high hit rate plus asymmetric magnitudes drive the Sharpe.²⁴

5.2.3 Discussion

The four clusters trace a single market cycle. In stages 1 and 2 (continuation), the strategy looks like classical momentum, with above-average winner selection in cluster 1 and a weakened version in cluster 2. In stages 3 and 4 (reversal), it looks nothing like classical momentum, inverting to loser selection in cluster 3 and to a term-structure-wide reversal in cluster 4. The regime-conditioning architecture is therefore not “momentum plus a regime control” but “momentum where momentum works, reversal where momentum crashes”.

XGB ranks stocks on predicted forward return rather than on a per-horizon momentum score, which allows the long leg to hold stocks not top-ranked at any individual horizon. The cluster analysis makes this concrete, with the long leg in cluster 4 sitting below the cross-sectional mean at every horizon from 1 to 12, and in cluster 3 sitting below the cross-sectional mean at every horizon even while the cross-sectional mean itself rallies. No single-horizon rank-based momentum strategy would select these stocks for the long side. The same compositional flexibility implements a regime-conditioned direction flip, classical winner selection (above-average z-scores) in calm and reversal (below-average z-scores) in panic, with the long-short portfolio fully invested throughout and the picks at panic peaks sitting uniformly below the cross-sectional mean across all twelve horizons.

The analysis makes two contributions to the momentum literature. First, the methodological combination is itself novel. Prior regime-conditional momentum work has paired simple regime treatments (binary up-or-down indicators, single-variable volatility thresholds, or fast-slow blending weights) with linear or rank-based cross-sectional rules. This thesis instead pairs a continuous probabilistic regime filter built from multiple stress indicators (the Bayesian HMM filtered probability π_t^{filter}) with a nonlinear cross-sectional model (XGBoost). The filtered probability acts as a routing variable through tree splits, conditioning the joint use of all twelve momentum horizons on the current market state.

The combination enables a class of cross-sectional selection not documented in the prior momentum literature. The selection can act on the full term-structure shape rather than per-horizon rank, can hold stocks not top-ranked at any individual horizon, and can take a different functional form across regimes. Classical (Jegadeesh and Titman, 1993), multi-horizon (Novy-Marx, 2012), and multi-asset (Asness et al., 2013) momentum strategies cannot produce these compositions by construction, and ML-based asset-pricing

²⁴Cluster-level leg-beta and SHAP-share diagnostics for the full $K = 4$ partition are in Appendix H.7.

methods that select on predicted return (Gu et al., 2020; Beckmeyer and Wiedemann, 2025) do not map their picks against the full term structure of past returns.

Second, the empirical analysis characterizes how the regime-momentum mechanism manifests at a granularity the prior literature does not provide. The fact that cross-sectional momentum profits flip sign across market states is itself documented descriptively in Cooper et al. (2004). What is new is the concrete form, a single nonlinear model routing between continuation- and reversal-mode selection on a continuous regime probability rather than a binary backward-looking state. The four-cluster decomposition above quantifies this routing at the z-score level, from roughly $+0.3$ to $+0.8\sigma$ above the cross-sectional mean in calm states to uniformly below it across all twelve horizons at panic peaks.

Existing crash-management approaches respond to panic instead by adjusting how much the strategy invests (Daniel and Moskowitz, 2016; Barroso and Santa-Clara, 2015), how the horizons are blended (Goulding et al., 2023), or by learning regime-agnostic signal reweightings (Beckmeyer and Wiedemann, 2025). XGB instead changes which stocks the strategy invests in, with the long-short portfolio fully invested. The timing of the response also differs: where GHM adapts to a more prominent short-horizon signal once trailing returns reveal a cycle change, this thesis identifies panic from real-time stress indicators and positions for the reversal at the onset of stress rather than after the cycle has shown up in prices.

Cluster-level CAPM betas confirm the architectural mechanism. XGB’s long leg holds high- β stocks in every cluster, including at panic peaks, so the strategy participates in market rebounds rather than being crushed by them (Appendix H.7, Table 27).

5.3 Robustness and Limitations

The strategy’s headline performance comes with three limitations. First, the baseline volatility is high (around 19.5%) and may require down-scaling for risk-averse investors. Section 5.3.1 traces the risk-return trade-off offered by a CRRA risk-aversion control. Second, the 2011–2024 test sample does not include the worst historical period for the proposed XGB strategy. Section 5.3.2 extends the evaluation to 25 years (2000–2024) via expanding-window backtests covering the dot-com bust, the GFC, and the 2009 momentum crash. Third, the panic-regime reversal rests on the premise that price declines during stress are temporary. A sustained economic deterioration would expose the strategy to reversal failure, which Section 5.3.3 discusses alongside the complementary D&M exposure-scaling overlay.

5.3.1 Risk-Return Trade-off

The first limitation is volatility itself. An investor may prefer lower portfolio risk at the cost of some expected return. We incorporate a risk-aversion parameter γ into the strategy and trace the out-of-sample trade-off across $\gamma \in [0, 1]$.

The natural starting point is the mean-variance utility $r - \frac{\gamma}{2}\sigma^2$ (Markowitz, 1952), but its penalty applies to absolute return variance. Empirical work on investor preferences finds that risk tolerance scales with wealth: a 10% drawdown is treated as a significant loss whether the portfolio is \$1M or \$100M, while a fixed \$100K loss is devastating for the smaller investor and trivial for the larger one. Mean-variance does not capture this scaling. Constant relative risk aversion (CRRA), $U(W) = W^{1-\gamma}/(1-\gamma)$, embeds proportional risk treatment by construction. Marginal utility declines at a rate that scales with wealth, so the implied penalty applies to wealth-normalized volatility rather than absolute return variance. We adopt CRRA as the risk-aversion control.

In practice, we make two changes to the naive mean-variance ranking. First, we use $\log \hat{\sigma}$ as the volatility penalty rather than $\hat{\sigma}^2$. The log scale makes the cost of risk proportional to its level, matching the wealth-scaling intuition above. Second, we score utility separately for each trading direction. Otherwise a rank-based long-short construction would pile low-volatility names into the long leg and high-volatility names into the short leg as γ rises, raising realized long-short volatility rather than lowering it. With per-direction scoring and the same volatility penalty applied in both directions, a volatile stock is equally unattractive whether considered as a holding or as a short position. For each stock s , the utility under each trading direction is

$$\begin{aligned} u_{\text{long}}^s &= +\text{sign}(\hat{r}^s) \log(1 + |\hat{r}^s|/\varepsilon) - \gamma \log(\hat{\sigma}^s), \\ u_{\text{short}}^s &= -\text{sign}(\hat{r}^s) \log(1 + |\hat{r}^s|/\varepsilon) - \gamma \log(\hat{\sigma}^s), \end{aligned} \tag{24}$$

where $\hat{r}^s$ is the XGBoost forward-return prediction, $\hat{\sigma}^s$ a forward-volatility forecast from a parallel XGBoost model trained on realized 12-month volatility, and $\varepsilon = 0.01$. Each month the model picks stocks that maximize these utilities. Longs are the top decile of u_{long} across the full cross-section, shorts are the top decile of u_{short} . Full specification, including the volatility-forecast model, is in Appendix F.

Risk-Aversion Sweep: Sharpe, Return, Volatility, Drawdown vs. γ

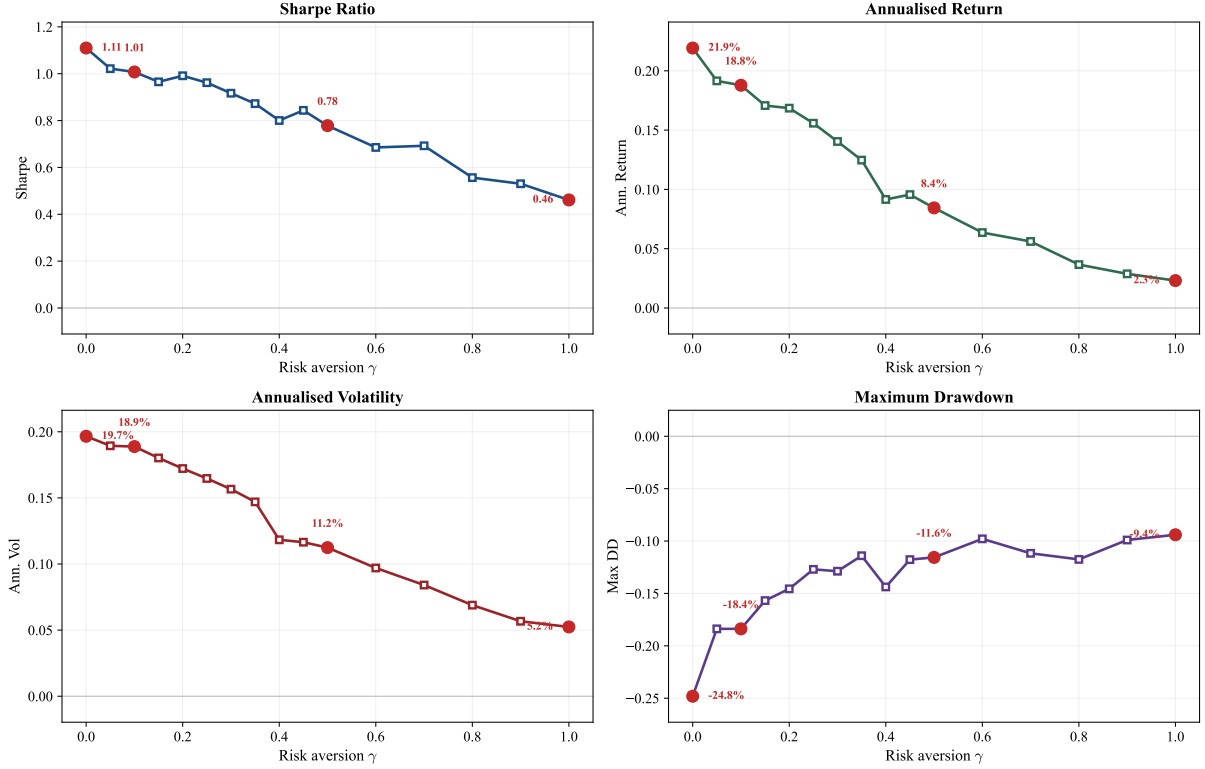


Figure 12: Risk-aversion sweep. Sharpe ratio, annualised return, annualised volatility, and maximum drawdown as a function of γ over $[0, 1]$. Red markers highlight $\gamma \in \{0, 0.1, 0.5, 1.0\}$.

Figure 12 traces the resulting risk-return trade-off as γ rises, where each point corresponds to a different γ and gives a different return-and-risk pair. At $\gamma = 0$ the volatility penalty vanishes and the utilities reduce to a monotonic transformation of the predicted return, so the picks approximately reproduce the headline XGB scoring (Sharpe 1.11; volatility 19.7% and MDD -25% versus 19.5% and -23% in Table 3, the small gap reflecting the per-direction utility transformation). At $\gamma = 0.1$, Sharpe stays at 1.01 while maximum drawdown tightens from -25% to -18%, a near-free reduction in tail risk. At $\gamma = 0.5$, annualised volatility halves from 19.7% to 11.2%, drawdown halves to -12%, and Sharpe settles at 0.78. At $\gamma = 1.0$, volatility falls to 5.2% (roughly a quarter of the market's) and Sharpe remains positive at 0.46. XGB's regime-conditional edge depends on taking volatility. Its reversal bet, especially cluster 4, loads the long leg with stocks that have fallen the furthest. These are typically the most volatile names in the cross-section, and the bet pays off when they rebound. As γ rises, the volatility penalty filters out exactly these names, and the strategy progressively loses the contrarian bet that generates its panic-regime Sharpe. Beyond $\gamma \approx 1.5$ the long-short spread collapses into dollar-neutral noise, comparable to the zero-Sharpe outcome of the linear baselines on the same features (Section 5.1).

5.3.2 Historical Stress: Out-of-Sample Evidence

The 14-year test sample used so far (2011–2024) misses the worst momentum-relevant stress events: the dot-com bust, the global financial crisis, and the 2009 momentum crash documented by [Daniel and Moskowitz \(2016\)](#). To extend the evaluation, the full pipeline (HMM + XGBoost) is re-estimated on an annual expanding-window schedule: annual retraining is initialized in 2000, with each year’s model trained only on data available through 31 December of the prior year and used to make predictions for the following year. Stitching these year-by-year predictions together produces a 25-year out-of-sample monthly return series spanning 2000–2024 (299 months).²⁵ Methodology details are in Appendix H.2.

Period	N months	XGB			Market		
		Sharpe	Total Return	Max DD	Sharpe	Total Return	Max DD
Full sample, 2000–2024	299	0.50	898.4%	−64.8%	0.56	572.6%	−50.3%
2000–2002 dot-com	36	−0.15	−35.9%	−64.8%	−0.72	−38.1%	−45.0%
2003–2006 bull	48	0.70	37.5%	−12.7%	1.70	80.5%	−4.6%
2007–2009 GFC	36	0.26	11.1%	−36.3%	−0.15	−13.9%	−50.3%
2010–2019 calm	120	0.67	155.4%	−17.9%	1.05	258.1%	−17.6%
2020–2024 COVID era	59	1.23	299.1%	−22.2%	0.81	95.2%	−24.8%

Table 9: Sub-period decomposition of the expanding-window out-of-sample backtest, 2000–2024. Annual retraining is initialised in 1995 using all data available through the prior December. Market is the value-weighted CRSP market (Mkt-RF + RF, Fama-French monthly factors).

Worst observed stress: the dot-com bust

The 2000–2002 dot-com bust produced the strategy’s largest historical drawdown of −64.8% peak-to-trough. Restricted to the bear-market proper (March 2000–October 2002), the strategy lost −43.8% cumulative over 32 months (Sharpe −0.39). This is the historical realization of the reversal-failure vulnerability discussed in Section 5.3.3 below: a prolonged sector-rotating bear in which beaten-down stocks did not promptly recover, so the panic-regime reversal selection bled.

Stress at the targeted failure mode: the 2009 momentum crash

The single most informative episode is March–December 2009, the period [Daniel and Moskowitz \(2016\)](#) document as a severe momentum crash. Under the same long-short construction used throughout this thesis (NYSE breakpoints, value-weighted, 10 bps cost), unconditional 12-month momentum lost −62.6% cumulative during those ten months. Under expanding-window out-of-sample, the regime-conditional strategy gained +20.5% cumulative (Sharpe 0.73) during the same window. Across the full GFC sub-window

²⁵The expanding-window construction enforces strict out-of-sample status throughout: each year’s predictions are produced by a model trained only on observations available at the corresponding cutoff, so no future information leaks into any prior-year prediction.

(2007–2009 in Table 9) the strategy returned +11.1% (Sharpe 0.26) while the market lost −13.9% (Sharpe −0.15). This is direct historical evidence that the regime mechanism survives the specific stress event that defeats both unconditional momentum and the broader market.

5.3.3 Reversal Failure and the D&M Complement

The dot-com episode above is the empirical realization of the strategy’s reversal-failure mode. The cluster 4 bet rests on the premise that panic-period declines are temporary, which holds in shorter panics; XGB earned +29.7% during the COVID crash (February–April 2020) and +20.8% during the 2022 bear market (January–October 2022), and in both cases the decline reversed within months. The premise fails in a sustained economic deterioration where beaten-down stocks do not promptly recover because their businesses are failing, as in the 32-month dot-com bear (March 2000–October 2002) where XGB lost −43.8% cumulative with a peak-to-trough drawdown of −64.8%. The regime classification was correct. What changed was the typical panic-regime pattern. Normally past losers in panic rebound. In the dot-com bust they did not, because the underlying businesses were failing.

XGB and Daniel and Moskowitz (2016) use regime information along complementary channels. XGB conditions *which* stocks the portfolio holds on the regime. D&M condition *how much* the portfolio holds on the regime. On the same data and transaction costs, XGB (Sharpe 1.11, CAPM alpha 23.4%) outperforms D&M managed momentum (Sharpe 0.49, CAPM alpha 10.3%). The two approaches are not substitutes. A hybrid that adds a D&M-style exposure-scaling overlay on top of XGB as a circuit breaker (Section 6.3) would reduce the risk of a total collapse during a prolonged bear market like the dot-com bust without sacrificing XGB’s stock-selection advantage in shorter panics.

5.3.4 International Robustness

To test whether the cross-sectional regime-momentum mechanism extends beyond US data, we replicate the full pipeline on UK and Japanese equities (Compustat Global, 1990–2024).²⁶

²⁶Each region’s HMM is fit on locally-selected stress features (details in Appendix K). The cross-sectional model is unchanged from the US specification, except that long and short legs are formed from unconditional decile breakpoints across all listed stocks, since no NYSE-equivalent tier exists for either market.

Strategy	UK				JP			
	Sharpe	Ann. Ret	Total Ret	Max DD	Sharpe	Ann. Ret	Total Ret	Max DD
Local market	0.52	5.5%	110.5%	−25.2%	0.76	10.7%	313.8%	−21.5%
Fixed 12-mo mom	0.35	6.1%	127.6%	−62.6%	−0.01	−1.9%	−23.0%	−66.3%
XGB	0.72	13.0%	449.2%	−21.4%	0.57	7.0%	155.1%	−19.8%

Table 10: International cross-sectional model performance, 2011–2024 OOS. Regionally-fitted HMM (DD+VOL+REL_N).

Both regions support the mechanism. UK XGB attains a Sharpe of 0.72 over 2011–2024, well above unconditional 12-month momentum’s 0.35. JP XGB attains 0.57, substantially above unconditional 12-month momentum’s −0.01, consistent with the documented weakness of Japanese momentum ([Asness et al., 2013](#)). Maximum drawdown is reduced from −62.6% (fixed momentum) to −21.4% in UK, and from −66.3% to −19.8% in JP.

XGB works in both regions. In the UK it dominates unconditional momentum on every dimension, with the same shape of improvement as the US. In Japan, where baseline cross-sectional momentum is essentially absent ([Asness et al., 2013](#)), XGB still extracts a Sharpe of 0.50. The regime conditioning is producing alpha that the unconditional momentum trade in Japan does not. XGB’s drawdowns cluster within four percentage points across all three markets (−22.8% US, −21.4% UK, −19.8% JP) against fixed-momentum drawdowns of −62% to −72%, so the regime conditioning compresses the panic tail to the same magnitude in every region. The HMM feature set was re-selected per region (Section 4), so what transfers is the mechanism, not a US-specific stress signal.

6 Conclusions

This thesis asked how cross-sectional momentum reorganizes across market regimes. The prior literature established that momentum is regime-dependent but did not specify the form that dependence takes. The two-stage framework developed here answers the question concretely: a continuous probabilistic regime signal routes a nonlinear cross-sectional model between classical winner selection in calm and term-structure-wide reversal at panic peaks.

6.1 Conclusions and Contributions

Cross-sectional momentum’s regime-dependent structure is real and admits a concrete characterization. The regime signal acts as a context variable, conditioning how a nonlinear cross-sectional model uses the full momentum term structure rather than predicting returns directly. The three-method ladder (DET, LR, XGB) introduced in Section 3 establishes that this conditioning is irreducibly nonlinear: the deterministic regime-adaptive formula and the linear classifier deliver marginal or negative Sharpes on the same features that XGB compounds into a Sharpe of 1.11, so nonlinearity is the limiting factor, not the conditioning variable itself. The analysis makes two contributions to the momentum literature, developed in Section 5.2.3. First, the methodological combination of a continuous probabilistic regime filter with a nonlinear cross-sectional model is novel relative to prior regime-conditional momentum work. It enables a class of cross-sectional selection not documented in the prior literature, including a regime-conditioned direction flip. The flip itself was noted descriptively by [Cooper et al. \(2004\)](#), but here it is implemented concretely as a single nonlinear model with the long-short portfolio fully invested throughout.

Second, the empirical analysis specifies how the cross-section reorganizes across regimes at the level of individual cluster z-scores, with the long leg sitting roughly $+0.3$ to $+0.8\sigma$ above the cross-sectional mean in calm and -0.7 to -0.9σ below at panic peaks. The regime signal acts as an irreducibly nonlinear routing variable for the momentum term structure. SHAP attributes 46% of importance to the regime signal even though it carries near-zero standalone return-predictive content, and removing it alone cuts the strategy’s Sharpe by more than 60%. The mechanism also extends beyond the main test sample, replicating on UK and Japanese equities with locally-fitted HMMs and surviving a 25-year expanding-window backtest covering the dot-com bust, the GFC, and the 2009 momentum crash.

6.2 Limitations

Reversal-failure vulnerability

The strategy’s primary vulnerability is a prolonged economic deterioration in which the prior decline reflects genuine business failure rather than temporary mispricing, so the beaten-down stocks do not recover. The dot-com bust is its historical realization (Section 5.3.2). A sustained multi-year decline without any reversal (a depression-style outcome) lies outside the 25-year historical sample.

Test period scope

The 2011–2024 test period does not contain a prolonged recession, and the headline 1.11 Sharpe rises steadily across sub-periods (0.59, 1.04, 1.70 across 2011–2015, 2016–2020, 2021–2024; Appendix G.1). Returns concentrate in the back half of the sample, where multiple distinct stress episodes (COVID, the 2022 inflation shock, the 2023 banking stress) drove π_t^{filter} high most often, activating the model’s panic-regime reversal behavior. The calm pre-2016 window sees π_t^{filter} rarely elevated, and the model earns more modest returns from standard winner continuation. A test window with a different mix of regime episodes would deliver a different point estimate.

Frozen HMM parameters and regime drift

The HMM is estimated on 1990–2010 and held fixed during the test period. This assumes that the statistical properties of calm and panic regimes remain stationary. If the market undergoes structural change (a permanent increase in baseline volatility due to algorithmic trading or shifting monetary policy regimes, for example), the trained thresholds may become miscalibrated, with structurally calm periods classified as panic. The sustained elevation of π_t^{filter} in recent years may partly reflect such drift rather than genuine prolonged stress. Online re-estimation, in which the HMM updates as new data arrives, would address this directly.

Specification search

The HMM feature set (DD+DISP+REL_N+CS) was selected in part by comparing candidates on out-of-sample portfolio Sharpe in Pass 2 of the four-pass procedure described in Section 4.1. This raises a standard data-snooping concern: when many feature specifications are evaluated on the same out-of-sample window, the winner’s reported Sharpe is upward-biased relative to what an investor with no foreknowledge would have achieved. Two design features mitigate this. Pass 1 screens candidates on MCMC convergence and crisis-window alignment, and Pass 4 filters on cross-seed stability; both operate on criteria independent of portfolio performance, so the Pass-2 Sharpe comparison selects among specifications that have already cleared two non-Sharpe gates. The leave-one-out variants in Table 13 produce qualitatively similar Sharpes, so the headline 1.11 should be read as the upper end of a tight band of similar specifications rather than as an isolated peak the

Pass-2 search picked out.

6.3 Recommendations for Future Research

Five directions extend this framework most directly.

Richer regime-signal representation

The current model uses only the contemporaneous filtered probability π_t^{filter} , discarding regime dynamics. Two complementary extensions follow. The first looks forward, computing a k -step regime forecast $P(s_{t+k} = 1 \mid \mathbf{z}_{1:t})$ from the HMM’s transition probabilities so the portfolio can reposition before a regime shift rather than after it. The gain is concentrated in transition months, where the current framework trades on a stale view of the regime. The second looks backward, feeding the trajectory of π_t^{filter} rather than its current value so that a fresh spike (where mean reversion is most likely) and a sustained elevation (where the reversal mechanism may be breaking down) become distinguishable. A sequential model, or simple lag-based feature engineering (lagged π_t^{filter} values, a short-window rate of change, a regime-duration counter), would expose this trajectory. Together, the two extensions provide anticipation and memory.

Alternative nonlinear models

XGBoost is one of several nonlinear architectures, and the central mechanism (that π_t^{filter} gates the use of momentum horizons) should in principle be recoverable by any sufficiently expressive model. The most natural alternative is a feedforward neural network, which [Gu et al. \(2020\)](#) find competitive with tree ensembles for cross-sectional return prediction, with performance peaking at three hidden layers. Training such a network on the same inputs as XGB (π_t^{filter} and the twelve momentum horizons) would test whether the regime-conditional shift is a property of the data, recoverable by any nonlinear model, or specific to XGBoost’s tree-split structure. Beyond the binary replication test, comparing the network’s per-feature attribution to the SHAP decomposition in [Table 7](#) would reveal whether the regime signal plays the same routing role in a non-tree architecture.

Short-leg structure

The empirical characterization in [Section 5](#) is built on long-leg cross-sectional z-curves; the short leg enters only by symmetric decile construction and is not characterized in its own right. [Appendix H.7](#) reports that the short basket is structurally distinct from a negated long leg, with short-vs-negated-long correlation collapsing at panic peaks (cluster 4). This indicates that the short leg follows different selection modes than the long leg, with [Nagel \(2012\)](#)’s short-horizon reversal mechanism a plausible driver of the cluster-4 divergence. Re-running the cluster decomposition directly on the short-leg z-curves, rather than relabeling long-leg clusters, would recover any short-side modes the current characterization misses. The analysis was deferred as beyond the scope of this thesis, since the parallel

decomposition would not have changed the headline mechanism documented from the long leg.

Regime-conditional feature sets

Section 5.1.3 shows that adding firm-level fundamentals to XGB hurts in panic while leaving calm performance roughly unchanged, suggesting the optimal feature set may itself be regime-dependent. A layered construction in which XGB produces the initial long-short ranking and fundamentals enter as a post-ranking filter (excluding bottom-quintile-quality names from the long leg, high-leverage names from the short leg) is a natural starting point. A more ambitious variant trains separate cross-sectional models for the calm and panic branches, each with its own feature set. Both designs require careful out-of-sample validation because the regime-conditional split reduces effective sample size.

Exposure-scaling overlay (D&M complement)

A D&M-style exposure-scaling overlay layered on top of XGB’s stock selection would provide a circuit breaker against sustained downturns that violate the panic-rebound premise the model assumes. The overlay requires a triggering rule; candidates include π_t^{filter} above a threshold for a minimum number of consecutive months, or its rolling maximum over a fixed window exceeding a calibrated bound. Preliminary checks on the 2011–2024 sample show no improvement, because the sample contains no prolonged recession and the overlay only reduces exposure during months that turn out to be profitable. Its value would emerge in a sustained downturn outside the current window; the dot-com sub-period of the 25-year expanding-window backtest (Section 5.3.2) is the appropriate setting to evaluate candidate overlay rules.

References

- James H. Albert and Siddhartha Chib. Bayes inference via Gibbs sampling of autoregressive time series subject to Markov mean and variance shifts. *Journal of Business & Economic Statistics*, 11(1):1–15, 1993. doi: 10.1080/07350015.1993.10509929.
- Clifford S. Asness, Tobias J. Moskowitz, and Lasse Heje Pedersen. Value and momentum everywhere. *The Journal of Finance*, 68(3):929–985, 2013. doi: 10.1111/jofi.12021.
- Nicholas Barberis, Andrei Shleifer, and Robert Vishny. A model of investor sentiment. *Journal of Financial Economics*, 49(3):307–343, 1998. doi: 10.1016/S0304-405X(98)00027-0.
- Pedro Barroso and Pedro Santa-Clara. Momentum has its moments. *Journal of Financial Economics*, 116(1):111–120, 2015. doi: 10.1016/j.jfineco.2014.11.010.
- Leonard E. Baum, Ted Petrie, George Soules, and Norman Weiss. A maximization technique occurring in the statistical analysis of probabilistic functions of Markov chains. *The Annals of Mathematical Statistics*, 41(1):164–171, 1970. doi: 10.1214/aoms/1177697196.
- Heiner Beckmeyer and Timo Wiedemann. All days are not created equal: Understanding momentum by learning to weight past returns. *Journal of Banking and Finance*, 181, 2025. doi: 10.1016/j.jbankfin.2025.107510.
- Daniele Bianchi, Andrea De Polis, and Ivan Petrella. Time-varying skewness and momentum crashes. *SSRN Working Paper*, 2022. doi: 10.2139/ssrn.4182040.
- Markus K. Brunnermeier and Lasse Heje Pedersen. Market liquidity and funding liquidity. *Review of Financial Studies*, 22(6):2201–2238, 2009. doi: 10.1093/rfs/hhn098.
- Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016. doi: 10.1145/2939672.2939785.
- Michael J. Cooper, Roberto C. Gutierrez Jr., and Allaudeen Hameed. Market states and momentum. *The Journal of Finance*, 59(3):1345–1365, 2004. doi: 10.1111/j.1540-6261.2004.00665.x.
- Kent Daniel and Tobias J. Moskowitz. Momentum crashes. *Journal of Financial Economics*, 122(2):221–247, 2016. doi: 10.1016/j.jfineco.2015.12.002.
- Werner F. M. De Bondt and Richard Thaler. Does the stock market overreact? *The Journal of Finance*, 40(3):793–805, 1985. doi: 10.1111/j.1540-6261.1985.tb05004.x.

- Arthur P. Dempster, Nan M. Laird, and Donald B. Rubin. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society: Series B (Methodological)*, 39(1):1–38, 1977. doi: 10.1111/j.2517-6161.1977.tb01600.x.
- Eugene F. Fama and Kenneth R. French. A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22, 2015. doi: 10.1016/j.jfineco.2014.10.010.
- Sylvia Frühwirth-Schnatter. *Finite Mixture and Markov Switching Models*. Springer, 2006. doi: 10.1007/978-0-387-35768-3.
- Andrew Gelman and Donald B. Rubin. Inference from iterative simulation using multiple sequences. *Statistical Science*, 7(4):457–472, 1992. doi: 10.1214/ss/1177011136.
- Andrew Gelman, John B. Carlin, Hal S. Stern, David B. Dunson, Aki Vehtari, and Donald B. Rubin. *Bayesian Data Analysis*. Chapman and Hall/CRC, 3rd edition, 2013. doi: 10.1201/b16018.
- John Geweke. Bayesian treatment of the independent student- t linear model. *Journal of Applied Econometrics*, 8:S19–S40, 1993. doi: 10.1002/jae.3950080504.
- Christian L. Goulding, Campbell R. Harvey, and Michele G. Mazzoleni. Momentum turning points. *Journal of Financial Economics*, 149(3):378–406, 2023. doi: 10.1016/j.jfineco.2023.05.011.
- Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273, 2020. doi: 10.1093/rfs/hhaa009.
- James D. Hamilton. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384, 1989. doi: 10.2307/1912559.
- Peter D. Hoff. *A First Course in Bayesian Statistical Methods*. Springer, 2009. doi: 10.1007/978-0-387-92407-6.
- Harrison Hong and Jeremy C. Stein. A unified theory of underreaction, momentum, and overreaction in asset markets. *The Journal of Finance*, 54(6):2143–2184, 1999. doi: 10.1111/0022-1082.00184.
- Harrison Hong, Terence Lim, and Jeremy C. Stein. Bad news travels slowly: Size, analyst coverage, and the profitability of momentum strategies. *The Journal of Finance*, 55(1):265–295, 2000. doi: 10.1111/0022-1082.00206.
- Narasimhan Jegadeesh. Evidence of predictable behavior of security returns. *The Journal of Finance*, 45(3):881–898, 1990. doi: 10.1111/j.1540-6261.1990.tb05110.x.

- Narasimhan Jegadeesh and Sheridan Titman. Returns to buying winners and selling losers: Implications for stock market efficiency. *The Journal of Finance*, 48(1):65–91, 1993. doi: 10.1111/j.1540-6261.1993.tb04702.x.
- Narasimhan Jegadeesh and Sheridan Titman. Profitability of momentum strategies: An evaluation of alternative explanations. *The Journal of Finance*, 56(2):699–720, 2001. doi: 10.1111/0022-1082.00342.
- Charles M.C. Lee and Bhaskaran Swaminathan. Price momentum and trading volume. *The Journal of Finance*, 55(5):2017–2069, 2000. doi: 10.1111/0022-1082.00280.
- Bruce N. Lehmann. Fads, martingales, and market efficiency. *The Quarterly Journal of Economics*, 105(1):1–28, 1990. doi: 10.2307/2937816.
- Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017.
- Scott M. Lundberg, Gabriel Erion, Hugh Chen, Alex DeGrave, Jordan M. Prutkin, Bala Nair, Ronit Katz, Jonathan Himmelfarb, Nisha Bansal, and Su-In Lee. From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, 2(1):56–67, 2020. doi: 10.1038/s42256-019-0138-9.
- Harry Markowitz. Portfolio selection. *The Journal of Finance*, 7(1):77–91, 1952. doi: 10.1111/j.1540-6261.1952.tb01525.x.
- Stefan Nagel. Evaporating liquidity. *The Review of Financial Studies*, 25(7):2005–2039, 2012. doi: 10.1093/rfs/hhs066.
- Robert Novy-Marx. Is momentum really momentum? *Journal of Financial Economics*, 103(3):429–453, 2012. doi: 10.1016/j.jfineco.2011.05.003.
- Yizhan Shu and John M. Mulvey. Dynamic factor allocation leveraging regime-switching signals. *arXiv preprint arXiv:2410.14841*, 2024.
- Tyler Shumway. The delisting bias in CRSP data. *The Journal of Finance*, 52(1):327–340, 1997. doi: 10.1111/j.1540-6261.1997.tb03818.x.
- Dimitri Vayanos and Paul Woolley. An institutional theory of momentum and reversal. *Review of Financial Studies*, 26(5):1087–1145, 2013. doi: 10.1093/rfs/hht014.

Appendix A: Data Sources and Variable Definitions

Three primary data sources are used. **CRSP** (Center for Research in Security Prices, via WRDS) provides monthly stock-level returns, prices, shares outstanding, exchange codes, and share codes from the Monthly Stock File (`crsp.msf`), as well as daily market returns from the Daily Stock Index file (`crsp.dsi`). **Compustat** (via WRDS) provides quarterly fundamental accounting data from `comp.fundq`, linked to CRSP via the CCM bridge table using primary link types (LU, LC). **FRED** (Federal Reserve Economic Data) provides Moody’s BAA and AAA corporate bond yields for computing the credit spread.

Following [Shumway \(1997\)](#), delisting returns are incorporated to prevent survivorship bias. When a stock is delisted, its actual delisting return is used if available; for performance-related delistings (CRSP codes 500–584) with missing returns, -30% is imputed.

Table 11 summarises all data variables.

Variable	Source	Description	Used in
<i>Stock-level variables</i>			
Monthly returns	CRSP <code>msf</code>	Adjusted holding-period returns	Sections 3, 5
Price, shares out.	CRSP <code>msf</code>	For market equity computation	Section 4
Exchange, share codes	CRSP <code>msf</code>	Universe filtering (codes 10, 11)	Section 4
Delisting returns	CRSP <code>msedelist</code>	Survivorship bias correction	Section 4
Book equity (<code>ceqq</code>)	Compustat <code>fundq</code>	Book-to-market, ROE	Appendix L
Total assets (<code>atq</code>)	Compustat <code>fundq</code>	Asset growth, gross profit, CF ratios	Appendix L
Income (<code>ibq</code>)	Compustat <code>fundq</code>	ROE, earnings growth, accruals	Appendix L
Revenue (<code>revtq</code>)	Compustat <code>fundq</code>	Gross profitability	Appendix L
COGS (<code>cogsq</code>)	Compustat <code>fundq</code>	Gross profitability	Appendix L
Debt (<code>dlttq</code> , <code>dltcq</code>)	Compustat <code>fundq</code>	Leverage ratio	Appendix L
Operating cash flow (<code>oancfy</code>)	Compustat <code>fundq</code>	CFO, FCF, accruals	Appendix L
Capital expenditure (<code>capxy</code>)	Compustat <code>fundq</code>	Free cash flow	Appendix L
<i>Market-level variables (HMM inputs)</i>			
Daily market returns	CRSP <code>dsi</code>	Drawdown (DD) computation	Section 4.2
Market price index	CRSP <code>dsi</code>	Drawdown (DD)	Section 4.2
Cross-sect. stock returns	CRSP <code>msf</code>	Dispersion (DISP), REL_N	Section 4.2
BAA, AAA bond yields	FRED	Credit spread (CS)	Section 4.2
<i>Derived features</i>			
$mom_1, \dots, mom_{12}$	Computed	12 trailing momentum signals	Section 4.4
π_t^{filter}	HMM output	Filtered panic probability	Section 4.3
10 fundamentals (ablation only)	Computed	bm, roe, gross profit, cash flow, etc.	Appendix L

Table 11: Summary of all data variables used in the analysis.

Notes: All data accessed via WRDS. Compustat linked to CRSP via the CCM bridge table using primary link types (LU, LC). Fundamentals are lagged four months from fiscal quarter end to ensure real-time availability. The sample covers January 1990 to November 2024 (training: 1990–2010, 240 months; test: 2011–2024, 167 months).

Appendix B: HMM Feature Selection Tables

The two tables in this section support the four-pass feature selection procedure described in Section 4.1. Table 12 reports the top 10 candidate feature combinations on ensemble Sharpe and cross-seed stability. Table 13 reports include-one and leave-one-out ablations of the production four-feature set.

D	Combination	Sharpe	Mean	σ
4	DD+DISP+REL_N+CS	0.91	0.84	0.11
4	DD+DISP+CS+TERM	0.82	0.55	0.15
4	DD+DISP+REL_N+SKEW	0.80	0.69	0.08
4	DD+VOL+DISP+REL_N	0.80	0.77	0.09
3	DD+DISP+REL_N	0.77	0.77	0.10
4	DD+VOL+DISP+TERM	0.71	0.67	0.16
3	DD+VOL+CS	0.69	0.43	0.13
1	DD	0.68	0.67	0.14
4	DD+DISP+REL_N+TERM	0.62	0.45	0.18
3	DD+ADR+TERM	0.57	0.69	0.07

Table 12: Top 10 feature combinations from the four-pass selection pipeline (long-short, 10 bps costs). D = number of HMM features. Sharpe = ensemble Sharpe ratio (10 HMM seeds, 20 XGBoost seeds). Mean and σ are the mean and standard deviation of the Sharpe ratio across 20 independent experiments with different seed partitions, measuring robustness to random initialisation. DD+DISP+REL_N+CS ranks first on both ensemble Sharpe and stability mean.

HMM Input Features	XGB Sharpe	Δ vs Full
Full 4F baseline (DD+DISP+REL_N+CS)	0.97	—
<i>Include-one (single-feature HMM)</i>		
DD only (market drawdown)	0.66	−0.31
<i>Leave-one-out (three-feature HMM)</i>		
Drop DD	0.51	−0.46
Drop DISP	0.96	−0.01
Drop REL_N	0.43	−0.54
Drop CS	0.96	−0.01

Table 13: HMM input feature ablation (long-short, feature-selection seed counts). The absolute Sharpe levels differ from the production model (Table 3) due to fewer seeds; the relative comparisons (Δ) are the relevant quantities. DD alone is the strongest single-feature classifier but trails the full set by −0.31. Removing DD or REL_N causes the largest leave-one-out drops (−0.46 and −0.54), while DISP and CS contribute marginally to the Sharpe but improve seed stability (Section 4.1).

Appendix C: Portfolio Formation, Transaction Costs, and Performance Metrics

C.1 Portfolio Construction

For each method, a long-short portfolio is constructed each month. Stocks are ranked by their scores, and NYSE-listed stocks are used to determine decile breakpoints. All stocks (including AMEX and NASDAQ) whose score exceeds the 90th percentile breakpoint enter the long leg; all stocks below the 10th percentile breakpoint enter the short leg. Long-short construction is the standard in the momentum literature (Jegadeesh and Titman, 1993; Daniel and Moskowitz, 2016; Barroso and Santa-Clara, 2015) because it isolates momentum’s cross-sectional stock selection from market beta, providing a cleaner test of the strategy’s ability to distinguish future winners from losers.

Portfolio weights within each leg are assigned proportionally to market equity (value-weighting):

$$w_t^{s,L} = \frac{ME_t^s}{\sum_{s \in \mathcal{L}_t} ME_t^s}, \quad w_t^{s,S} = \frac{ME_t^s}{\sum_{s \in \mathcal{S}_t} ME_t^s}, \quad (25)$$

where $\mathcal{L}_t$ and $\mathcal{S}_t$ are the sets of stocks in the long and short legs at month t , respectively. The long-short portfolio return is:

$$r_t^{LS} = \sum_{s \in \mathcal{L}_t} w_t^{s,L} \cdot r_t^s - \sum_{s \in \mathcal{S}_t} w_t^{s,S} \cdot r_t^s. \quad (26)$$

Value-weighting tilts each leg toward large, liquid stocks, reducing the influence of small-cap return anomalies and improving implementability.

C.2 Transaction Costs

All portfolios are rebalanced monthly.

A one-way transaction cost of $c = 10$ basis points is applied to the turnover in both the long and short legs at each rebalancing date. The net portfolio return at month t is:

$$r_t^{\text{net}} = r_t^{\text{gross}} - c \cdot (\tau_t^L + \tau_t^S), \quad (27)$$

where τ_t^L and τ_t^S are the one-way turnover in the long and short legs, respectively:

$$\tau_t^L = \frac{1}{2} \sum_s \left| w_t^{s,L,\text{new}} - w_{t-1}^{s,L,\text{prev}} \right|, \quad \tau_t^S = \frac{1}{2} \sum_s \left| w_t^{s,S,\text{new}} - w_{t-1}^{s,S,\text{prev}} \right|, \quad (28)$$

and each summation runs over all stocks appearing in either the new or previous portfolio for that leg. The 10 basis point cost reflects the transaction environment for a value-

weighted, large-cap-biased strategy where NYSE breakpoints tilt toward liquid securities.

Transaction costs are applied *post-hoc* as an accounting adjustment and are not incorporated into the models’ training objectives. The cross-sectional models (DET, LR, and XGB) optimize stock-level prediction (classification accuracy or return forecasting) without knowledge of the previous portfolio, the resulting turnover, or the cost of rebalancing. This is standard practice in the empirical asset pricing literature (e.g., [Fama and French, 2015](#); [Goulding et al., 2023](#)), where portfolio construction is treated as a fixed downstream step applied uniformly to all strategies. An end-to-end framework that jointly optimizes stock selection and portfolio-level objectives (e.g., net-of-cost Sharpe ratio) would require a fundamentally different architecture, such as a reinforcement learning or differentiable portfolio optimization layer, and remains a potential extension of this work.

C.3 Performance Metrics

Standard Metrics

Strategy performance is assessed using four standard risk-adjusted metrics computed on monthly net-of-fee returns.

Annualized return The geometric annualized return over n months is:

$$R_{\text{ann}} = \left[\prod_{t=1}^n (1 + r_t) \right]^{12/n} - 1. \quad (29)$$

Annualized volatility The annualized standard deviation of monthly returns:

$$\sigma_{\text{ann}} = \sigma(r) \cdot \sqrt{12}. \quad (30)$$

Sharpe ratio The annualized Sharpe ratio:²⁷

$$SR = \frac{\bar{r}}{\sigma(r)} \cdot \sqrt{12}, \quad (31)$$

where $\bar{r}$ is the mean monthly return. Sharpe is computed from the arithmetic monthly mean annualised by $\sqrt{12}$, while annualised returns reported elsewhere in this thesis are compound (geometric); under high volatility the two definitions of average return differ, so Sharpe should not be inferred from a table by dividing annualised return by annualised volatility (e.g. at depth 1 in the methodology hyperparameter sweep, the reported Sharpe is 0.53 but $8.7\%/19.4\% = 0.45$).

Maximum drawdown Let $C_t = \prod_{\tau=1}^t (1 + r_\tau)$ denote the cumulative wealth path.

²⁷All Sharpe ratios use raw returns. The risk-free rate averaged $\sim 2\%$ over 2011–2024 (near zero pre-2021), so absolute Sharpe ratios are overstated by roughly 0.05–0.10; relative comparisons are unaffected because the convention is uniform.

The maximum drawdown is:

$$MDD = \min_{1 \leq t \leq n} \frac{C_t - \max_{1 \leq \tau \leq t} C_\tau}{\max_{1 \leq \tau \leq t} C_\tau}. \quad (32)$$

Regime-Conditional Evaluation

To assess whether the regime signal adds value beyond unconditional stock selection, performance is decomposed by market regime. Months are partitioned into calm ($\pi_t^{\text{filter}} < 0.5$) and panic ($\pi_t^{\text{filter}} \geq 0.5$) periods, and the Sharpe ratio is computed separately within each subset:

$$SR_{\text{calm}} = \frac{\bar{r}_{\text{calm}}}{\sigma(r_{\text{calm}})} \cdot \sqrt{12}, \quad SR_{\text{panic}} = \frac{\bar{r}_{\text{panic}}}{\sigma(r_{\text{panic}})} \cdot \sqrt{12}. \quad (33)$$

This decomposition helps assess whether the strategy’s alpha originates primarily from calm-period stock selection, panic-period risk avoidance, or both.

Per-Cluster Sharpe

The cluster Sharpes reported in Section 5.2.2 apply the standard Sharpe formula to the months in each cluster. Let M_C denote the set of months assigned to cluster C , let r_t be the long-short monthly portfolio return at month t , and let $n_C = |M_C|$. The point-estimate cluster Sharpe is

$$\text{Sharpe}_C = \frac{\bar{r}_{M_C}}{\sigma(r_{M_C})} \cdot \sqrt{12},$$

where

$$\bar{r}_{M_C} = \frac{1}{n_C} \sum_{t \in M_C} r_t, \quad \sigma(r_{M_C}) = \sqrt{\frac{1}{n_C - 1} \sum_{t \in M_C} (r_t - \bar{r}_{M_C})^2}.$$

The denominator is therefore the unbiased sample standard deviation of the in-cluster long-short monthly returns; annualisation by $\sqrt{12}$ follows the monthly-to-annual convention used throughout this thesis. Confidence intervals come from a block bootstrap (block size 6, 5,000 reps) on the in-cluster return series.

Appendix D: Feature Importance Methods

D.1 SHAP Values for XGBoost

To interpret the XGBoost model and assess the marginal contribution of each feature, SHAP values (Lundberg and Lee, 2017) are computed on the test set. The SHAP value of feature j for observation $\mathbf{x}$ measures its average marginal contribution to the prediction across all possible feature orderings:

$$\phi_j(\mathbf{x}) = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|! (p - |S| - 1)!}{p!} [f_{S \cup \{j\}}(\mathbf{x}) - f_S(\mathbf{x})], \quad (34)$$

where N is the full feature set, $p = |N|$, and $f_S(\mathbf{x})$ is the expected model output when only features in S are observed. For tree-based models, the TreeSHAP algorithm (Lundberg et al., 2020) computes these values exactly in polynomial time. The overall importance of each feature is summarized as:

$$I_j = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} |\phi_j(\mathbf{x}_i)|. \quad (35)$$

Two additional analyses are performed. First, the mean absolute SHAP value of π_t^{filter} is computed separately for calm and panic months to assess whether the regime signal’s influence varies across market states. Second, regime-specific XGBoost models are trained on calm-only and panic-only subsets of the training data, and their SHAP feature importances are compared to identify which features the model relies on differentially across regimes.

Appendix E: Alternative Training Targets for XGBoost

Section 5.3.1 argues that embedding a risk penalty directly in the XGBoost training target degrades performance, motivating the CRRA specification used in the main text. Table 14 shows this degradation is not an artefact of the mean-variance functional form: the same pattern appears under three families of training targets with varying risk tolerance.

Target family	Parameter	Sharpe	Ann. Ret	Ann. Vol	MDD
Baseline (r)	–	1.107	30.0%	27.1%	–28.3%
Mean-variance: $r - \frac{\gamma}{2}\sigma^2$	$\gamma = 0.5$	1.061	22.9%	21.6%	–22.3%
	$\gamma = 1$	1.024	16.5%	16.2%	–22.8%
	$\gamma = 2$	1.135	15.5%	13.6%	–22.0%
	$\gamma = 5$	1.062	13.5%	12.7%	–22.5%
	$\gamma = 10$	1.082	13.1%	12.1%	–18.9%
Log mean-variance: $\log(1+r) - \frac{\gamma}{2}\sigma^2$	$\gamma = 0.5$	1.051	14.6%	13.9%	–17.9%
	$\gamma = 1$	1.039	13.8%	13.4%	–23.8%
	$\gamma = 2$	1.080	14.0%	12.9%	–20.9%
	$\gamma = 5$	1.127	14.1%	12.4%	–18.7%
	$\gamma = 10$	0.995	12.1%	12.3%	–20.2%
Sharpe-like: r / σ^a	$a = 0.5$	1.148	24.8%	21.4%	–22.4%
	$a = 1.0$	0.984	14.6%	15.0%	–29.1%
	$a = 1.5$	0.936	12.4%	13.5%	–21.0%
	$a = 2.0$	0.759	12.8%	18.1%	–24.5%

Table 14: Out-of-sample performance of XGBoost (XGB) under alternative training targets. The construction here is long-only (top-decile, NYSE-breakpoint, value-weighted), used as a diagnostic that isolates the effect of the training target itself from the long-short construction; the absolute Sharpes are therefore not comparable with the long-short headline in Table 3, only the relative ordering across rows. Every risk-adjusted target reduces returns faster than volatility.

The pattern is consistent: returns fall faster than volatility under every risk penalty, regardless of functional form. The mildest adjustments (mean-variance with $\gamma = 0.5$, Sharpe-like with $a = 0.5$) come closest to the baseline but still underperform. This confirms that the diagnostic in Section 5.3.1 is not specific to the mean-variance utility formulation; the single-model training-target approach is structurally unsuited to incorporating risk aversion in a long-short setting, which the long-and-short CRRA construction resolves.

Appendix F: Long and Short CRRA Utilities

This appendix supplements the long-and-short utility specification in Section 5.3.1 with the volatility-forecast model, the high- γ collapse mechanism, and the sweep grid.

Volatility forecast

The expected-return forecast $\hat{r}_t^s$ comes from the XGBoost ensemble described in Section 3. The volatility forecast $\hat{\sigma}_t^s$ comes from a second XGBoost ensemble trained on the realized 12-month forward volatility

$$\sigma_{[t+1, t+12]}^s = \text{std}(r_{t+1}^s, \dots, r_{t+12}^s),$$

using the same feature set augmented with trailing 12-month realized volatility. The 50-seed ensemble is trained on 1990–2010 and evaluated out-of-sample on 2011–2024.

Convergence to dollar-neutral noise at high γ

Stocks appearing in both top deciles of u_{long} and u_{short} (the overlap set, which grows with γ) are retained on both sides, and their contributions to the long and short books net out in $r_L - r_S$. This is the mechanism by which the long-short spread converges to zero as γ rises beyond the useful range. At high γ , the volatility penalty dominates the return signal, low- $\hat{\sigma}$ stocks occupy both top deciles simultaneously, and the two legs become near-identical value-weighted low-volatility portfolios.

Sweep grid

The risk-aversion sweep in Figure 12 uses a 17-point grid over $[0, 1]$: step 0.05 up to 0.5, step 0.1 thereafter.

Appendix G: Robustness Checks

This section reports six robustness checks that complement the main results in Section 5.

G.1 Sub-Period Stability

To assess whether the main findings are driven by a single favorable sub-period, the test sample is split into three roughly equal windows: 2011–2015, 2016–2020, and 2021–2024.

	2011–2015	2016–2020	2021–2024	Full
Market	0.72	1.04	0.73	0.84
Fixed 12-mo	0.74	-0.20	-0.11	0.06
LR	0.47	-0.02	-0.38	-0.01
XGB	0.59	1.04	1.70	1.11

Table 15: Sub-period Sharpe ratios (long-short). The test period is split into three roughly equal sub-periods. XGB is the only strategy with positive Sharpe ratios in all three sub-periods.

In long-short construction, XGB is the only strategy that produces positive Sharpe ratios in every sub-period (0.59, 1.04, 1.70). LR collapses in 2016–2020 (Sharpe -0.02) and turns negative in 2021–2024 (-0.38). Fixed 12-month momentum turns negative after 2015, consistent with the momentum crash mechanism documented by Daniel and Moskowitz (2016). XGB’s strongest performance (1.70) is in 2021–2024, a period characterized by elevated volatility and multiple stress episodes, precisely the conditions where regime-conditional stock selection adds the most value.

G.2 Turnover and Transaction Cost Sensitivity

A strategy’s net-of-cost performance depends critically on its turnover. Table 16 reports average monthly one-way turnover for each strategy.

Strategy	Avg. Monthly TO	Annualised TO
Fixed 12-mo	70.1%	842%
Fixed 1-mo	184.3%	2,211%
LR	168.8%	2,026%
XGB	132.3%	1,588%

Table 16: Portfolio turnover by strategy (long-short). Average monthly one-way turnover and annualised turnover (monthly $\times$ 12).

In long-short construction, XGB’s average monthly one-way turnover is 132.3%, reflecting the regime-conditional changes in stock selection that shift the portfolio composition substantially. Table 17 translates these turnover differences into Sharpe ratio sensitivity across a range of transaction cost assumptions.

	0 bps	5 bps	10 bps	20 bps	30 bps	50 bps
Fixed 12-mo	0.10	0.08	0.06	0.03	0.00	-0.06
Fixed 1-mo	0.39	0.32	0.26	0.14	0.01	-0.24
LR	0.12	0.05	-0.02	-0.16	-0.29	-0.57
XGB	1.19	1.15	1.11	1.03	0.95	0.78

Table 17: XGB remains profitable at 50 bps. All other strategies become unprofitable at moderate cost levels.

XGB remains profitable across all cost levels tested: even at 50 bps one-way (five times the baseline), its Sharpe ratio is 0.78. All other strategies become unprofitable at moderate cost levels. The baseline 10 bps assumption is reasonable for a value-weighted strategy using NYSE breakpoints, which tilts toward the most liquid names.

G.3 Model Sensitivity

The XGBoost model uses hyperparameters selected via cross-validation on the training set. To verify that the results are not an artefact of a particular configuration, Table 18 reports out-of-sample performance under alternative tree depths, learning rates, and ensemble sizes.

Configuration	Sharpe	Ann. Ret	Ann. Vol
depth=3, lr=0.05, $n=500$	0.98	19.2%	19.9%
depth=4, lr=0.05, $n=500$	1.11	21.7%	19.5%
depth=5, lr=0.05, $n=500$	0.92	17.4%	19.4%
depth=6, lr=0.05, $n=500$	0.90	17.0%	19.5%
depth=4, lr=0.01, $n=500$	0.99	18.7%	19.2%
depth=4, lr=0.10, $n=500$	1.00	19.6%	19.9%
depth=4, lr=0.05, $n=200$	1.06	20.5%	19.4%
depth=4, lr=0.05, $n=1000$	0.96	18.7%	19.9%

Table 18: XGBoost hyperparameter sensitivity (long-short). Each row varies one hyperparameter from the baseline. The baseline is not the single best configuration, but all variants outperform unconditional momentum.

Hyperparameter sensitivity

At the baseline depth of 4, the Sharpe ratio ranges from 0.89 to 1.11 across learning rate and ensemble size variations. Performance degrades at depths below 3 (see Section 3.2.1 for the depth analysis) and above 5 due to overfitting. The results confirm that XGB’s performance is not an artefact of a particular hyperparameter configuration.

G.4 Number of Hidden States

The baseline HMM uses $K = 2$ states (calm and panic). A natural question is whether finer regime granularity (for example, distinguishing mild corrections from severe crises) would improve the downstream portfolio models. Table 19 reports out-of-sample Sharpe ratios for $K = 2, 3, 4, 5$, each averaged over five representative Gibbs sampler seeds (the production regime signal averages 200 seeds, but five suffice for this diagnostic).

K	LR		XGB	
	Mean Sharpe	Std	Mean Sharpe	Std
2	-0.012	0.000	1.107	0.000
3	-0.029	0.004	0.831	0.164
4	-0.029	0.003	0.606	0.291
5	-0.028	0.003	0.724	0.087

Table 19: Out-of-sample Sharpe ratios for HMMs with $K = 2, 3, 4, 5$ states (long-short). The $K=2$ row uses the 200-seed production regime signal as the reference point (its Std is zero by construction). For $K = 3, 4, 5$ each entry reports the mean and standard deviation across 5 independent Gibbs sampler seeds (2,000 iterations each). LR is invariant to K . XGB shows no consistent improvement beyond $K = 2$, and cross-seed variability increases with K .

In long-short construction, XGB shows no consistent improvement beyond $K = 2$. Cross-seed variability increases with K , and the intermediate states that emerge for $K \geq 3$

lack the persistence needed to provide a stable conditioning signal. With only 240 training months, the data cannot reliably separate more than two regimes for the purpose of downstream cross-sectional stock selection.

Beyond estimation, $K = 2$ has an interpretability advantage. The two-state model produces a single continuous signal (π_t^{filter}) that XGBoost can condition on in a straightforward way: higher values indicate more stress. With $K \geq 3$, the additional states may capture dimensions of the economy that are orthogonal to the continuation-reversal mechanism, for example a “growth” state characterized by strong fundamentals but moderate stress indicators. Such a state would conflate two economically distinct situations (a genuine expansion vs. a recovery from panic) and could introduce noise into the regime signal, degrading the model’s ability to distinguish continuation from reversal. The two-state framework avoids this by restricting the regime classification to a single dimension, the degree of market stress, which is the dimension most relevant to momentum’s cross-sectional structure.

G.5 January Exclusion

Momentum is known to reverse in January (Jegadeesh and Titman, 1993): past losers outperform past winners, partially offsetting momentum’s annual profitability. To verify that XGB’s performance is not driven by a January seasonal, Table 20 recomputes all metrics after excluding the 14 January months from the test period. XGB’s Sharpe ratio declines marginally from 1.11 to 1.06, confirming that the regime-momentum interaction operates throughout the calendar year. Fixed 12-month momentum improves slightly when January is excluded (0.06 to 0.14), consistent with the well-documented January reversal in unconditional momentum.

	Ann. Ret	Ann. Vol	Sharpe	Months
<i>Full sample (baseline)</i>				
XGB	21.7%	19.5%	1.11	167
Fixed 12-mo mom	−1.9%	26.1%	0.06	167
<i>Excluding January</i>				
XGB	20.9%	19.9%	1.06	153
Fixed 12-mo mom	0.1%	25.6%	0.14	153

Table 20: January exclusion test. Performance is recomputed after dropping all 14 January months from the test period.

Appendix H: External Validity

H.1 Baseline Train/Test Split

Table 21 reports the baseline train/test split performance. Robustness across training windows and prediction horizons is provided separately by the 25-year expanding-window backtest (Appendix H.2 below, also reported in Section 5.3.2), which re-estimates the full pipeline annually and therefore subsumes a continuum of alternative splits.

Train / Test split	N_{test}	LR Sharpe	XGB Sharpe
1990–2010 / 2011–2024 (baseline)	167	−0.01	1.11

Table 21: Out-of-sample performance under the baseline train/test split (long-short, mom+ π , 50-seed ensemble). LR collapses while XGB achieves a Sharpe of 1.11 with highly significant factor model alphas.

In the baseline split (1990–2010 training, 2011–2024 test), XGB achieves a Sharpe of 1.11 while LR collapses (−0.01), confirming the nonlinear regime-momentum interaction documented in Section 5.1.

H.2 25-Year Expanding-Window Out-of-Sample Methodology

The historical evidence reported in Section 5.3.2 comes from a separate analysis from the one above: a 25-year out-of-sample backtest in which the full pipeline (HMM + XGBoost) is re-estimated every January, with annual retraining initialized in 2000 and each year’s predictions made by a model trained only on data available through the prior December. Each retraining uses the production seed configuration described in Section 3 (200 HMM seeds, 50 XGBoost seeds) and the same long-short construction described in Appendix C.1.

The HMM’s panic-state label is fixed by an economically motivated sign convention rather than the data-driven percentile comparison used elsewhere. The convention encodes the known direction of each feature in panic regimes: −1 for drawdown, +1 for cross-sectional dispersion, −1 for relative market participation, and +1 for the credit spread. The data-driven label-switching procedure used in the main pipeline is unstable on short training windows with few observed crisis months (under 20), so the fixed-sign fallback activates for retrains before 2002. From 2002 onward the data-driven procedure resumes, by which point the training window contains the dot-com bust as a calibration anchor. The HMM seed loop is parallelised across CPU cores via `multiprocessing.Pool`; the parallel implementation is bit-identical to the serial reference (verified in the test suite by comparing per-seed pi-filter arrays).

Each retraining produces 12 monthly out-of-sample long-short returns, which are stitched into a single monthly return series; this thesis reports the 25-year sub-sample from January 2000 to November 2024 (299 months). Every prediction is strictly causal: the model used for month t saw no information after month $t - 1$. The 167-month sub-window from January 2011 onward overlaps the main test period reported in Section 5.1 and serves as a methodological cross-check; expanding-window OOS produces a Sharpe of 0.88 over that window, within the production result’s bootstrap interval.

H.3 Factor Model Regressions

To test whether XGB’s out-of-sample return is explained by known risk factors, monthly long-short returns are regressed on five standard factor models: CAPM, Fama–French three-factor, Carhart four-factor, Fama–French five-factor, and a six-factor specification that augments FF5 with the momentum factor (UMD). Each regression takes the form

$$r_t^{LS} = \alpha + \boldsymbol{\beta}^\top \mathbf{f}_t + \varepsilon_t, \quad (36)$$

where $\mathbf{f}_t$ is the factor return vector for the specification in question, $\boldsymbol{\beta}$ measures the strategy’s exposure to each factor, and α is the average return earned in excess of what those exposures would mechanically generate. The null hypothesis $\alpha = 0$ states that the factor model fully explains the strategy’s returns; rejecting it means the strategy earns return that no linear combination of the factors can replicate. Newey–West standard errors (6 lags) are used throughout to correct for residual autocorrelation and heteroskedasticity in monthly returns. Table 22 reports $\hat{\alpha}$ and its t -statistic under each specification; Table 23 repeats the exercise for the fund-augmented variant. Alphas that survive all five factor models indicate return that cannot be attributed to market, size, value, profitability, investment, or momentum exposure.

Model	α (%)	$t(\alpha)$	Mkt-RF	SMB	HML	UMD	RMW	CMA
CAPM	23.4	4.08	-0.14					
FF3	23.2	4.48	-0.14	+0.04	-0.17			
Carhart	24.3	4.75	-0.19	-0.01	-0.22	-0.20		
FF5	22.9	4.62	-0.13	+0.08	-0.23		+0.06	+0.12
FF6	24.1	4.81	-0.18	+0.01	-0.31	-0.22	+0.01	+0.21

Table 22: Factor model regressions for XGB (mom+ π). α is annualised. t -statistics use Newey–West standard errors (6 lags).

For completeness, Table 23 reports the same regressions for the fund-augmented variant (XGB: mom+ π +fund) discussed in Section 5.1.3. Alphas remain statistically significant under all five specifications ($t > 2.9$) but are materially smaller than the baseline’s, consistent with the dilution documented in the main text.

Model	α (%)	$t(\alpha)$	Mkt-RF	SMB	HML	UMD	RMW	CMA
CAPM	17.6	3.04	-0.16					
FF3	17.1	3.12	-0.14	-0.06	-0.13			
Carhart	18.2	3.22	-0.19	-0.12	-0.18	-0.20		
FF5	16.4	3.02	-0.11	+0.01	-0.30		+0.09	+0.35
FF6	17.7	3.17	-0.16	-0.06	-0.39	-0.23	+0.04	+0.44

Table 23: Factor model regressions for XGB (mom+ π +fund). α is annualised. Newey–West t -statistics (6 lags).

H.4 Block Bootstrap Inference

To assess the sampling variability of XGB’s Sharpe ratio and test its outperformance formally, a block bootstrap is applied to the monthly return series. Returns are resampled with replacement in 12-month blocks to preserve autocorrelation, producing 10,000 synthetic 167-month paths (fixed seed 42). The Sharpe ratio is recomputed on each resample to obtain a 95% confidence interval, and the same block draws are used for XGB and each benchmark to form a paired test of the Sharpe difference. The bottom panel repeats the exercise on XGB returns split by regime.

	Sharpe	95% CI	vs XGB (p)
XGB	1.11	[0.67, 1.53]	—
LR	-0.01	[-0.41, 0.42]	0.001***
DET (Formula)	0.11	[-0.36, 0.59]	0.008***
Fixed 12-mo mom	0.06	[-0.43, 0.59]	0.005***
Fixed 1-mo mom	0.26	[-0.21, 0.73]	0.021**
<i>XGB regime-conditional Sharpe</i>			
XGB Calm ($n=108$)	0.84	[0.29, 1.32]	—
XGB Panic ($n=59$)	1.54	[0.85, 2.22]	—
Panic – Calm	0.70	[-0.16, 1.59]	0.109

Table 24: Block bootstrap confidence intervals for Sharpe ratios and paired tests of XGB’s outperformance. Uses 12-month blocks and 10,000 resamples with a fixed seed. The “vs XGB (p)” column reports two-sided p -values from paired tests where both strategies are resampled on the same dates. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

XGB’s Sharpe confidence interval is [0.67, 1.53], comfortably above zero, while every benchmark’s interval straddles zero. The paired tests reject equality of Sharpe ratios between XGB and each benchmark at $p < 0.025$. Within XGB, the panic-period Sharpe exceeds the calm-period Sharpe in point estimate (1.53 vs 0.84), but with only 59 panic months the paired difference is not statistically significant ($p = 0.109$); the 95% interval [-0.16, 1.59] straddles zero. The point-estimate gap is consistent with the regime-conditional selection mechanism described in Section 5.2, but its statistical confirmation would require a longer panic sample. These intervals describe sampling variability within

the 2011–2024 sample; out-of-sample generalization is addressed separately by the 25-year expanding-window backtest (Section 5.3.2; methodology in Appendix H.2) and the sub-period analysis (Appendix G.1).

H.5 Leg-Level CAPM Betas by Regime

The D&M comparison in Section 5.3.3 characterizes XGB’s contribution as stock selection rather than exposure scaling. A direct test of this framing is to compute leg-level CAPM betas separately by regime: if XGB reorganizes which stocks it holds across regimes (cross-sectional re-ranking), the long-leg and short-leg betas should shift between calm and panic in opposite directions; if instead XGB mechanically scales exposure, both legs’ betas should move together.

Table 25: Leg-level CAPM betas by regime, 2011–2024 OOS test period. Panic months are those with $\pi_t^{\text{filter}} \geq 0.5$. Newey–West standard errors (3 lags) in parentheses.

Strategy	Leg	Full	Calm	Panic
Fixed 12-mo mom	Long	1.09 (0.06)	1.09 (0.09)	1.07 (0.08)
Fixed 12-mo mom	Short	1.74 (0.11)	1.80 (0.21)	1.69 (0.13)
XGB	Long	1.55 (0.08)	1.40 (0.11)	1.65 (0.10)
XGB	Short	1.10 (0.06)	1.22 (0.09)	0.98 (0.05)

The pattern in Table 25 is consistent with the cross-sectional re-ranking interpretation. XGB’s long-leg beta rises from 1.40 in calm to 1.65 in panic (*tilt amplification*: in panic, the long leg holds names with higher market sensitivity), while the short-leg beta falls from 1.22 to 0.98. Fixed 12-month momentum, by contrast, holds both leg-level betas roughly stable across regimes (long: 1.09 calm, 1.07 panic; short: 1.80 calm, 1.69 panic), consistent with the unconditional rank-based construction. The difference is direct evidence that the regime signal acts on *which* stocks are held rather than on *how much*.

H.6 Per-cluster Feature Signature ($K = 4$ Partition)

Section 5.2.2 reports a $K = 4$ partition of the 167 test months and characterizes each cluster narratively. Table 26 provides the full per-cluster feature signature for completeness: cross-section context features (regime composition, momentum landscape, dispersion, skewness, recent strategy performance) computed at the month level and averaged across each cluster’s months; long-leg pick raw momentum at each horizon, averaged across each cluster’s months; and long-leg pick cross-sectional z-score (deviation from the cross-section mean at each horizon, in units of cross-section std) averaged across cluster months. Per-cluster Sharpes and other strategy outcomes appear in Table 8 in the main text.

Feature	Cluster 1 <i>calm cont.</i>	Cluster 2 <i>mild cont.</i>	Cluster 3 <i>post-panic</i>	Cluster 4 <i>deep crisis</i>
<i>Cross-section context</i>				
Cross-section mom (overall mean)	13.8%	7.3%	13.5%	−9.3%
Cross-section mom (short tertile, $h = 1-4$)	6.1%	4.4%	3.0%	−7.1%
Cross-section mom (mid tertile, $h = 5-8$)	14.6%	7.6%	12.4%	−11.1%
Cross-section mom (long tertile, $h = 9-12$)	20.8%	9.7%	25.0%	−9.6%
Cross-section dispersion (short)	0.23	0.27	0.28	0.23
Cross-section dispersion (mid)	0.44	0.46	0.56	0.35
Cross-section dispersion (long)	0.60	0.59	0.85	0.45
Cross-section skewness (short)	4.6	7.2	6.4	3.7
Cross-section skewness (mid)	6.2	7.6	6.2	5.6
Cross-section skewness (long)	6.7	8.1	6.5	5.7
π^{filter} frequency, prior 6 months	0.27	0.37	0.30	0.51
π^{filter} frequency, prior 12 months	0.31	0.32	0.41	0.30
Strategy Sharpe, prior 12 months	1.22	1.04	1.15	1.08
<i>Long-leg pick raw momentum by horizon</i>				
Long-leg pick mom, $h = 1$	7.1%	3.1%	−3.4%	−15.3%
Long-leg pick mom, $h = 2$	13.5%	8.6%	−5.5%	−21.6%
Long-leg pick mom, $h = 3$	18.3%	8.8%	−5.3%	−26.3%
Long-leg pick mom, $h = 4$	23.1%	10.6%	−5.8%	−30.8%
Long-leg pick mom, $h = 5$	30.8%	13.8%	−5.9%	−35.0%
Long-leg pick mom, $h = 6$	40.2%	18.0%	−3.2%	−38.3%
Long-leg pick mom, $h = 7$	49.4%	20.7%	−1.3%	−42.1%
Long-leg pick mom, $h = 8$	59.2%	23.7%	0.0%	−44.9%
Long-leg pick mom, $h = 9$	61.3%	23.1%	1.1%	−46.5%
Long-leg pick mom, $h = 10$	60.7%	21.9%	1.3%	−47.2%
Long-leg pick mom, $h = 11$	61.1%	22.8%	1.2%	−47.3%
Long-leg pick mom, $h = 12$	57.9%	18.5%	−5.3%	−47.5%
<i>Long-leg pick cross-sectional z-score by horizon</i>				
Long-leg z-curve, $h = 1$	0.26	0.04	−0.28	−0.73
Long-leg z-curve, $h = 2$	0.38	0.14	−0.31	−0.74
Long-leg z-curve, $h = 3$	0.40	0.11	−0.34	−0.75
Long-leg z-curve, $h = 4$	0.44	0.10	−0.36	−0.80
Long-leg z-curve, $h = 5$	0.54	0.16	−0.35	−0.85
Long-leg z-curve, $h = 6$	0.65	0.23	−0.29	−0.83
Long-leg z-curve, $h = 7$	0.73	0.27	−0.26	−0.85
Long-leg z-curve, $h = 8$	0.80	0.32	−0.22	−0.87
Long-leg z-curve, $h = 9$	0.76	0.29	−0.23	−0.88
Long-leg z-curve, $h = 10$	0.72	0.24	−0.25	−0.87
Long-leg z-curve, $h = 11$	0.67	0.22	−0.26	−0.86
Long-leg z-curve, $h = 12$	0.54	0.12	−0.35	−0.87

Table 26: Per-cluster feature signature for the $K = 4$ partition, 2011–2024 test period. Features are grouped into three blocks: (1) cross-section context features (regime composition, momentum landscape, dispersion, skewness, recent strategy performance) computed at the month level and averaged across each cluster’s months; (2) long-leg pick raw momentum at each horizon, averaged across each cluster’s months; (3) long-leg pick cross-sectional z-score (deviation from the cross-section mean at each horizon, in units of cross-section std) averaged across cluster months. Per-cluster Sharpes and other strategy outcomes appear in Table 8; this table is the full descriptive feature signature.

H.7 Cluster-Level Cross-Validation

Three diagnostics extend the regime-aggregate findings of Section 5.2 across the $K = 4$ partition.

Leg-betas across the $K = 4$ partition

XGB retains long-leg $\beta >$ short-leg β in every cluster (spreads +0.68, +0.40, +0.49, +0.43 for clusters 1–4), while unconditional 12-month momentum exhibits leg-beta inversion in every cluster (−0.24, −0.87, −0.75, −0.42; Table 27). Daniel and Moskowitz (2016)’s crash mechanism is latent across every cluster type; XGB escapes it everywhere through compositional re-ranking.

Cluster	n	XGB			Fixed 12-mo mom		
		β_L	β_S	Spread	β_L	β_S	Spread
1 (calm continuation)	53	1.65 (0.22)	0.97 (0.13)	0.68	1.32 (0.22)	1.57 (0.24)	−0.24
2 (transition)	62	1.60 (0.15)	1.20 (0.09)	0.40	1.04 (0.08)	1.91 (0.17)	−0.87
3 (post-panic recovery)	31	1.46 (0.16)	0.97 (0.11)	0.49	1.09 (0.17)	1.84 (0.22)	−0.75
4 (deep crisis)	21	1.49 (0.08)	1.06 (0.09)	0.43	1.07 (0.10)	1.49 (0.14)	−0.42

Table 27: Leg-level CAPM betas by K=4 cluster, 2011–2024 test period. XGB (regime-conditional XGBoost ensemble) versus unconditional 12-month momentum, side-by-side. Standard errors in parentheses. Spread = $\beta_L - \beta_S$.

π_t^{filter} ’s SHAP share by cluster

Combined long-and-short SHAP shares are 0.50, 0.46, 0.54, 0.24 for clusters 1–4 (Table 28). The pattern is consistent with the routing interpretation: π_t^{filter} contributes most where the cross-section is ambiguous (cluster 3) and least where it is unambiguously crisis-shaped (cluster 4, where momentum splits alone suffice to produce the loser portfolio).

Cluster	n	Long π share	Short π share	Combined π share
1 (calm continuation)	53	34.0%	50.1%	49.9%
2 (transition)	62	26.9%	46.5%	46.0%
3 (post-panic recovery)	31	27.8%	53.7%	53.6%
4 (deep crisis)	21	11.4%	38.1%	24.4%

Table 28: π_t^{filter} ’s SHAP share by K=4 cluster, 2011–2024 test period. Long and short shares are computed over long-leg and short-leg picks respectively; combined is over all stock-months in the cluster.

Short-leg structure (future work)

The $K = 4$ partition was defined on long-leg z-curves; applying the same labels to the short leg shows the short basket is structurally distinct from a negated long-leg, with short-vs-negated-long correlations of +0.94, +0.89, +0.19, and −0.84 for clusters 1–4 (Table 29). The month-1 short-leg spike in cluster 4 (+0.57) is consistent with Nagel

(2012)’s finding that short-horizon reversal returns spike during high-volatility periods: month 1 carries 10.6% of panic short-leg momentum SHAP versus 5.3% in calm. A full short-leg analysis is left to future work.

Cluster	n	z_1	$\bar{z}$	Basket	$\rho(-z^L)$
1 (calm continuation)	53	−0.062	−0.365	391	0.943
2 (transition)	62	0.101	−0.174	334	0.888
3 (post-panic recovery)	31	0.209	0.241	382	0.190
4 (deep crisis)	21	0.568	0.170	353	−0.835

Table 29: Short-leg z-curve summary by K=4 cluster, 2011–2024 test period. z_1 is the cluster-mean cross-sectional z-score of short-leg picks at the 1-month horizon; $\bar{z}$ averages across all 12 horizons. Basket size is mean stocks per month. $\rho(-z^L)$ is the correlation between the short-leg z-curve centroid and the negated long-leg z-curve centroid; values below 0.95 indicate structural asymmetry.

H.8 Emission Distribution Robustness

To verify that regime identification is robust to the choice of emission distribution, the HMM is re-estimated with multivariate Student- t emissions across a grid of degrees of freedom $\nu \in \{3, 5, 7, 10, 15, 20, 30, 50, 100\}$.

The Student- t model is estimated via data augmentation (Geweke, 1993): each observation receives a latent weight $w_t \sim \text{Gamma}(\nu/2, \nu/2)$, so that $\mathbf{z}_t \mid s_t=k, w_t \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k/w_t)$. Marginalising over w_t yields a multivariate t with ν degrees of freedom. Conditional on w_t the emission is Gaussian, so the Normal-Inverse-Wishart conjugate updates carry over with weighted sufficient statistics, preserving the efficiency of the Gibbs sampler.

ν	LL (train)	LL (test)	BIC	Agreement	Corr(π)
3	−882.7	−701.1	1929.9	91.4%	0.872
5	−829.0	−669.8	1822.5	96.8%	0.960
7	−838.3	−698.2	1841.2	98.0%	0.984
10	−815.8	−644.3	1796.1	96.8%	0.960
15	−800.1	−651.0	1764.8	98.3%	0.997
20	−802.9	−621.7	1770.4	99.3%	0.997
30	−802.6	−640.6	1769.8	98.5%	0.993
50	−796.6	−658.1	1757.7	98.8%	0.996
100	−792.3	−643.5	1749.0	98.8%	0.987
Normal	−787.5	−642.5	1739.5	—	—

Table 30: Student- t vs. Normal emission HMM comparison. LL = log-likelihood; Agreement = fraction of months with identical binary regime classification as the Normal HMM; Corr(π) = correlation of filtered panic probabilities.

The Normal model achieves the lowest BIC (2257.3 vs. 2257.6 for the best Student- t at

$\nu = 100$), indicating no meaningful improvement from heavier tails. Regime classifications agree with the Normal HMM on at least 97.8% of months across all ν values, and the correlation of filtered panic probabilities exceeds 0.98 everywhere.

Appendix I: Label Sign Correction

Because the four HMM features respond to stress in different directions, the expected direction of each feature during stress is determined empirically. For each feature j , the 95th percentile is computed separately on training months falling within known crisis windows (the dot-com bust of 2000–2002 and the Global Financial Crisis of 2007–2009) and on all remaining training months:

$$\text{sign}_j = \begin{cases} +1 & \text{if } q_{0.95}^{\text{crisis}}(z_j) \geq q_{0.95}^{\text{normal}}(z_j), \\ -1 & \text{otherwise.} \end{cases} \quad (37)$$

Features that spike upward during crises receive $\text{sign}_j = +1$; features that decline receive $\text{sign}_j = -1$. The panic state is then identified as the regime $k \in \{0, 1\}$ whose posterior mean vector $\hat{\mu}_k$ best aligns with the crisis direction:

$$k^* = \arg \max_{k \in \{0, 1\}} \sum_{j=1}^D \text{sign}_j \cdot \hat{\mu}_{k,j}. \quad (38)$$

The state labeled k^* is designated as “panic” and the other as “calm.”

Appendix J: Convergence Diagnostics

Convergence of the Gibbs sampler is assessed in two complementary ways: effective sample sizes (ESS) within each chain, and the Gelman–Rubin $\hat{R}$ statistic across independent chains. The two are complementary because ESS measures efficiency within a single chain but cannot detect whether different chains have converged to different posterior modes, while Gelman–Rubin compares chains but says nothing about within-chain autocorrelation. Trace plots provide a third, visual check.

J.1 Effective Sample Size (ESS)

Because successive MCMC draws are autocorrelated, n draws from the chain contain less information than n independent samples. The ESS quantifies the number of truly independent draws the chain is equivalent to. For a post-burn-in chain of length n ,

$$ESS = \frac{n}{1 + 2 \sum_{\ell=1}^{L^*} \hat{\rho}(\ell)}, \quad (39)$$

where $\hat{\rho}(\ell)$ is the estimated autocorrelation at lag ℓ ,²⁸ and L^* is the first lag at which the autocorrelation becomes negative. High ESS indicates that the chain provides many effectively independent posterior draws; an ESS of at least 100 per parameter is a common minimum for reliable posterior inference (Gelman et al., 2013).

²⁸ $\hat{\rho}(\ell) = \frac{1}{(n-\ell)\hat{\sigma}^2} \sum_{i=1}^{n-\ell} (\theta_i - \bar{\theta})(\theta_{i+\ell} - \bar{\theta})$, where θ_i is the i -th post-burn-in draw, $\bar{\theta}$ its sample mean, and $\hat{\sigma}^2$ its sample variance.

J.2 Trace plots

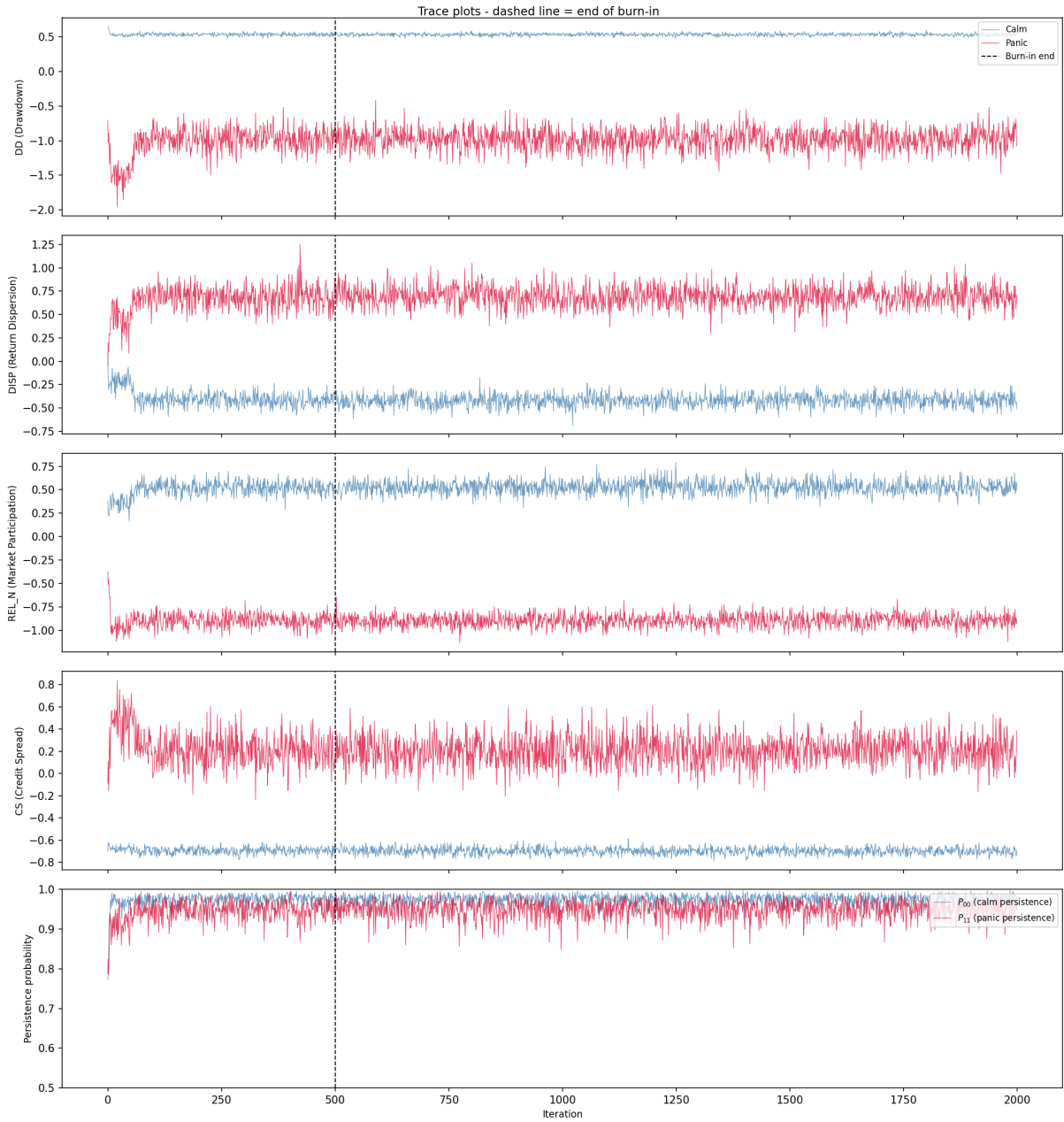


Figure 13: Trace plots of the Gibbs sampler. Each panel shows sampled regime means (calm in blue, panic in red) for one HMM feature. The dashed line marks the end of burn-in.

J.3 Gelman–Rubin Diagnostic

The Gelman–Rubin $\hat{R}$ statistic (Gelman and Rubin, 1992; Gelman et al., 2013) compares within-chain and between-chain variance across M independent chains, each of length n .

Let s_m^2 denote the variance of chain m and $\bar{\theta}_m$ its mean. The within-chain variance is

$$W = \frac{1}{M} \sum_{m=1}^M s_m^2, \quad (40)$$

and the between-chain variance is

$$B = \frac{n}{M-1} \sum_{m=1}^M (\bar{\theta}_m - \bar{\theta})^2, \quad (41)$$

where $\bar{\theta} = \frac{1}{M} \sum_m \bar{\theta}_m$. If all chains have converged to the same posterior, B should be small relative to W . The diagnostic combines these:

$$\hat{R} = \sqrt{\frac{\hat{V}}{W}}, \quad \hat{V} = \frac{n-1}{n} W + \frac{B}{n}. \quad (42)$$

Values of $\hat{R}$ near 1.0 indicate convergence; the conventional threshold is $\hat{R} < 1.1$, with a stricter modern bound at $\hat{R} < 1.01$.

Parameter	$\hat{R}$	Status
$\mu_{\text{panic}}(\text{DD})$	1.000	OK
$\mu_{\text{panic}}(\text{DISP})$	1.000	OK
$\mu_{\text{panic}}(\text{REL_N})$	1.000	OK
$\mu_{\text{panic}}(\text{CS})$	1.000	OK
$\mu_{\text{calm}}(\text{DD})$	1.000	OK
$\mu_{\text{calm}}(\text{DISP})$	1.001	OK
$\mu_{\text{calm}}(\text{REL_N})$	1.000	OK
$\mu_{\text{calm}}(\text{CS})$	1.000	OK
P_{00} (calm persistence)	1.000	OK
P_{01} (calm $\rightarrow$ panic)	1.000	OK
P_{10} (panic $\rightarrow$ calm)	1.000	OK
P_{11} (panic persistence)	1.000	OK

Table 31: Gelman–Rubin $\hat{R}$ statistics for all HMM parameters, computed across 5 independent chains (1,500 post-burn-in draws each). All values are below 1.01, well within the convergence threshold of 1.1.

J.4 Seed Convergence of the Regime Signal

Gelman–Rubin and the trace plots assess whether a single Gibbs chain has converged to its stationary posterior, but they do not quantify how many chains are needed to stabilize the *downstream* Sharpe ratio. The regime signal π_t^{filter} enters the pipeline as the average of N_s seed-specific forward filters (Section 3.1.5), and because XGBoost is a nonlinear function of π_t^{filter} , MCMC noise in any individual chain does not cancel linearly.

To measure this dependence, we subsample $k \in \{1, 2, 3, 5, 10, 15, 20, 30, 50, 75, 100\}$

chains without replacement from the seed pool, average the forward filter over those k chains, retrain the XGBoost ensemble on the resulting signal, and record the out-of-sample Sharpe. For each $k < 100$ we repeat on 30 random subsets; $k = 100$ exhausts the pool.

k	n_{sub}	Mean	Median	Std	P25	P75	Min	Max
1	30	1.040	1.031	0.078	0.985	1.090	0.915	1.205
2	30	1.062	1.081	0.069	1.019	1.103	0.894	1.217
3	30	1.049	1.044	0.043	1.014	1.075	0.973	1.143
5	30	1.042	1.042	0.047	1.002	1.083	0.967	1.154
10	30	1.053	1.050	0.036	1.038	1.066	0.965	1.145
15	30	1.075	1.079	0.029	1.055	1.097	1.008	1.139
20	30	1.078	1.080	0.022	1.065	1.090	1.025	1.122
30	30	1.089	1.082	0.025	1.069	1.109	1.052	1.137
50	30	1.093	1.092	0.015	1.082	1.101	1.067	1.124
75	30	1.098	1.102	0.012	1.091	1.105	1.067	1.118
100	1	1.118	1.118	0.000	1.118	1.118	1.118	1.118

Table 32: HMM Gibbs-chain seed convergence of the downstream L/S Sharpe ratio. For each subset size k , we draw n_{sub} random subsets of k HMM seeds from a 100-seed pool, average their forward-filter outputs into π_t^{filter} , retrain the XGBoost ensemble on the averaged signal, and report the out-of-sample L/S Sharpe. The production configuration uses $N_s = 200$ HMM seeds; the 100-seed pool here is the subset over which the experiment is run. Mean Sharpe stabilizes around $k \approx 30$ –50; standard deviation across subsets shrinks as $1/\sqrt{k}$.

Two patterns emerge. First, the mean Sharpe rises from 1.04 at $k = 1$ to 1.12 at $k = 100$, so a single-chain point estimate understates the strategy’s performance by roughly 0.08 in Sharpe units. The gap closes approximately monotonically (with minor non-monotonic dips at $k = 3$ and $k = 5$ before the trend resumes), and most of the lift is realized by $k = 30$. Second, the interquartile range across random subsets collapses from 0.10 at $k = 1$ to 0.02 at $k = 50$: once $k \geq 50$, any two random subsets of the seed pool produce nearly the same XGBoost portfolio. The production configuration ($N_s = 200$) sits well inside this plateau, which is why the 1.11 Sharpe reported in Section 5 is insensitive to which specific seeds were drawn. The practical implication is methodological rather than cosmetic: a single-chain HMM would not reliably support the main conclusions, so the seed ensemble is a requirement of the design, not a late-stage performance tuning.

Appendix K: International HMM Feature Selection

For the international robustness check (Section 5.3.4), the HMM feature set is selected per region using the same 4-pass procedure as the US production specification (Section 4.1): a quality screen on MCMC convergence and crisis-window alignment, a portfolio-Sharpe ranking on the surviving combinations, a definitive validation at higher

seed counts, and an additional stability check across random seed partitions. The international candidate pool comprises the five locally-available stress features (DD, VOL, DISP, REL_N, BANK_REL); BANK_REL is the value-weighted bank-sector excess return over the broad regional market, sign-flipped so high values indicate stress, and substitutes for the US-specific Moody’s BAA–AAA spread. With DD required, the procedure evaluates fifteen combinations of size one to four. Both UK and Japan select $DD+VOL+REL_N$ as the regional feature set. The transplanted-from-US template (DD+DISP+REL_N+BANK_REL) fails the Pass-1 quality screen for both regions (minimum effective sample size of 14 and 4 respectively), motivating per-region selection. Pass-by-pass CSV results and chosen-features JSON manifests are saved under `results/thesis/` with the prefix `intl_<region>_hmm`.

Appendix L: Fundamental Feature Definitions

Ten firm-level characteristics are computed from Compustat quarterly data (`fundq`), linked to CRSP via the CCM bridge table. All fundamentals are lagged four months from fiscal quarter end to ensure real-time availability. Cash-flow items (`oancfy`, `capxy`) are reported year-to-date and are differenced within each fiscal year to recover the per-quarter contribution before the trailing twelve-month sum is taken.

- **Book-to-market (bm):** Book equity (`ceqq`) divided by market equity.
- **Return on equity (roe):** Income before extraordinary items (`ibq`) divided by book equity.
- **Gross profitability:** Revenue minus cost of goods sold, scaled by total assets: $(\text{revtq} - \text{cogsq})/\text{atq}$.
- **Asset growth:** Year-over-year growth in total assets: $\text{atq}_t/\text{atq}_{t-4} - 1$.
- **Leverage:** Total debt divided by total assets: $(\text{dlttq} + \text{dlcq})/\text{atq}$.
- **Earnings growth:** Year-over-year growth in earnings, transformed as $\text{sign}(g) \cdot \log(1 + |g|)$.
- **Log market equity:** Natural logarithm of market capitalisation (price $\times$ shares outstanding), lagged one month.
- **Operating cash flow over assets (cfo_a):** Trailing twelve-month operating cash flow (`oancfy`, converted to quarterly and summed) scaled by total assets.
- **Free cash flow over assets (fcf_a):** Trailing twelve-month operating cash flow minus trailing twelve-month capital expenditure (`capxy`), scaled by total assets.

- **Accruals:** The accrual component of earnings, $(\text{ibq}_{\text{TTM}} - \text{oancf}_{\text{TTM}})/\text{atq}$, which captures the non-cash portion of reported earnings.

Fundamentals are winsorized at the 1st and 99th percentiles (5th and 95th for asset growth) using training-sample bounds. The cash-flow features (cfo_a , fcf_a , accruals) are available for roughly 83% of stock-months; the remaining seven characteristics exceed 82% coverage individually, and XGBoost handles any residual missingness natively via its default-branch logic.